Emergent Mathematics

Paul Gustafson

Contents

Part I

Symmetry

1 Hamiltonian Geometry

Definition 1.1 (Linear spaces and maps)**.** Let k be a field. A *linear space* over k is an abelian group V with scalar multiplication satisfying

$$a(v+w) = av + aw, \quad (a+b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map $T : V \to W$ is *linear* when $T(av + bw) = aT(v) + bT(w)$. Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For $T : V \to V$, a scalar λ is an *eigenvalue* when $\ker(T - \lambda I) \neq 0$; a nonzero vector in this kernel is an *eigenvector*.

Definition 1.2 (Dual spaces and tensor products)**.** The *dual space* of V is $V^\vee = \operatorname{Hom}_k(V, k)$. Its canonical evaluation pairing is

$$\mathrm{ev}_V : V^\vee \times V \longrightarrow k, \qquad (\varphi, v) \longmapsto \varphi(v).$$

The algebraic transpose of $T : V \to W$ is $T^\vee : W^\vee \to V^\vee$, defined by $T^\vee(\varphi) = \varphi \circ T$.

The *tensor product* $V \otimes W$ is characterized by a bilinear map $(v, w) \mapsto v \otimes w$ with the following universal property: every bilinear map $b : V \times W \to X$ factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\widetilde{b}} X.$$

The evaluation pairing induces a linear map $\mathrm{ev}_V : V^\vee \otimes V \to k$.

Definition 1.3 (Finite-dimensional linear spaces)**.** A linear space V is *finite-dimensional* when there is a coevaluation map

$$\mathrm{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the snake identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^\vee}) \circ (\mathrm{id}_{V^\vee} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^\vee}.$$

Definition 1.4 (Exterior powers, dimension, and determinant)**.** For $m \geq 0$, the mth *exterior power* of a linear space V is

$$\Lambda^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m : v_i = v_j \text{ for some } i \neq j \rangle,$$

and the image of $v_1 \otimes \cdots \otimes v_m$ is written $v_1 \wedge \cdots \wedge v_m$. An endomorphism T induces

$$\Lambda^m T(v_1 \wedge \cdots \wedge v_m) = Tv_1 \wedge \cdots \wedge Tv_m.$$

The products

$$\Lambda^r V \otimes \Lambda^s V \longrightarrow \Lambda^{r+s} V, \qquad (u_1 \wedge \cdots \wedge u_r) \otimes (v_1 \wedge \cdots \wedge v_s) \longmapsto u_1 \wedge \cdots \wedge u_r \wedge v_1 \wedge \cdots \wedge v_s$$

make $\Lambda^\bullet V = \bigoplus_{m\geq 0} \Lambda^m V$ the *exterior algebra* of V. A finite-dimensional linear space V has *dimension* n when $\Lambda^n V \neq 0$ and $\Lambda^{n+1} V = 0$. Its *determinant* is the scalar $\det(T)$ defined by the action on the highest nonzero exterior power:

$$\Lambda^n T = \det(T)\,\mathrm{id}_{\Lambda^n V}\,.$$

Definition 1.5 (Trace)**.** For a finite-dimensional linear space V, let $s_{V,V^\vee} : V \otimes V^\vee \to V^\vee \otimes V$ exchange the factors. For $T \in \mathrm{End}(V)$, define

$$\mathrm{tr}(T) = \mathrm{ev}_V \circ s_{V,V^\vee} \circ (T \otimes \mathrm{id}_{V^\vee}) \circ \mathrm{coev}_V\,.$$

Exercise 1.1 (Dimension, trace, and determinant)**.** Prove that every finite-dimensional linear space has a unique dimension n and that

$$\mathrm{tr}(\mathrm{id}_V) = n\,1_k.$$

Prove

$$\dim \Lambda^m V = \binom{n}{m}, \qquad \Lambda^m V = 0 \quad (m > n).$$

Prove that the exterior algebra is graded-commutative:

$$\xi \wedge \eta = (-1)^{rs} \eta \wedge \xi, \qquad \xi \in \Lambda^r V, \quad \eta \in \Lambda^s V.$$

Use functoriality of the highest exterior power to prove

$$\det(ST) = \det(S)\det(T), \qquad T \text{ is invertible} \iff \det(T) \neq 0.$$

Definition 1.6 (Smooth manifolds and tangent spaces)**.** A *smooth manifold* of dimension n is a Hausdorff, second-countable space with a maximal atlas of homeomorphisms to open subsets of $\mathbb{R}^n$ whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

The *tangent space* T_pM is the linear space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of a smooth map $F : M \to N$ is

$$dF_p : T_pM \longrightarrow T_{F(p)}N, \qquad dF_p(v)(f) = v(f \circ F).$$

Definition 1.7 (Tangent and cotangent bundles)**.** Define

$$TM = \coprod_{p\in M} T_pM, \qquad T^*M = \coprod_{p\in M} T_p^*M.$$

A smooth vector bundle is a smooth map $\pi : E \to M$ locally isomorphic to $U \times W \to U$ for a finite-dimensional real linear space W, by fiberwise linear maps. A section is a smooth map $s : M \to E$ satisfying $\pi s = \mathrm{id}_M$. The bundles TM and T^*M have the bundle structures induced by the differentials of chart maps and their duals. A classical position–momentum state is a pair (q,p) with $p \in T_q^*M$; for $\hbar > 0$, its wave covector is $p/\hbar$.

Definition 1.8 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. Its local flow is a family of local diffeomorphisms Φ_t satisfying

$$\Phi_0 = \mathrm{id}_M, \qquad \Phi_{s+t} = \Phi_s \circ \Phi_t, \qquad \frac{d}{dt} f(\Phi_t(p)) = X_{\Phi_t(p)}(f)$$

where both sides are defined. For vector fields X, Y, define

$$[X,Y](f) = X(Yf) - Y(Xf).$$

Definition 1.9 (Differential forms)**.** A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$; write $\Omega^k(M)$ for the space of differential k-forms. The exterior products on the cotangent spaces define

$$\wedge : \Omega^r(M) \times \Omega^s(M) \longrightarrow \Omega^{r+s}(M).$$

For a smooth map $F : M \to N$, define

$$(F^*\alpha)_p(v_1, \ldots, v_k) = \alpha_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

For a vector field X, define contraction by

$$(\iota_X \alpha)_p(v_2, \ldots, v_k) = \alpha_p(X_p, v_2, \ldots, v_k).$$

The *exterior derivative* is the family $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and $F^*d = dF^*$. For a vector field X with local flow Φ_t, define

$$\mathcal{L}_X \alpha = \left.\frac{d}{dt}\right|_{t=0} \Phi_t^* \alpha.$$

Exercise 1.2 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

Definition 1.10 (Cotangent phase space)**.** For the projection $\pi : T^*Q \to Q$, define the tautological one-form by

$$\theta_\alpha(v) = \alpha(d\pi(v)), \qquad \alpha \in T^*Q, \quad v \in T_\alpha(T^*Q).$$

The canonical two-form on T^*Q is $\omega = d\theta$.

Exercise 1.3 (From linear to cotangent phase space)**.** For a finite-dimensional real linear space V, construct the canonical identifications

$$TV \cong V \times V, \qquad T^*V \cong V \times V^*.$$

Prove that a phase $S \in C^\infty(Q)$ determines the momentum $p = dS_q \in T_q^*Q$ and wave covector $p/\hbar$. For a diffeomorphism $F : Q \to Q$, construct its cotangent lift and prove that it preserves θ and $d\theta$. Under $T^*V \cong V \times V^*$, prove

$$\omega((q,p),(q',p')) = p(q') - p'(q).$$

Definition 1.11 (Symplectic manifolds)**.** A *symplectic form* on a smooth manifold M is a closed nondegenerate two-form ω. A *symplectic linear space* is a finite-dimensional real linear space with a nondegenerate alternating bilinear form, regarded as a constant symplectic form. For $f \in C^\infty(M)$, define

$$\iota_{X_f}\omega = -df, \qquad \{f, g\} = -\omega(X_f, X_g).$$

The local flow of X_f is the *Hamiltonian flow* of f. A diffeomorphism $F : (M, \omega_M) \to (N, \omega_N)$ is a *canonical transformation* when $F^*\omega_N = \omega_M$.

Exercise 1.4 (Darboux theorem)**.** Prove that the canonical two-form $d\theta$ on T^*Q is nondegenerate and hence symplectic. Prove that every symplectic linear space of dimension $2n$ is symplectically isomorphic to

$$(\mathbb{R}^{2n}, \omega_0), \qquad \omega_0((q,p),(q',p')) = p(q') - p'(q).$$

Prove that every point of a $2n$-dimensional symplectic manifold has a neighborhood symplectomorphic to an open subset of $(\mathbb{R}^{2n}, \omega_0)$. Prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. For the Hamiltonian flow Φ_t of H, prove

$$\frac{d}{dt} f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

Exercise 1.5 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the volume form $\omega^n/n!$, and every density depending only on the Hamiltonian. On a symplectic linear space, prove that the infinitesimal generator of a linear Hamiltonian flow has trace zero and that the flow has determinant one.

Exercise 1.6 (Elliptic pendulum period)**.** On T^*S^1 consider

$$H(\theta, p) = \frac{p^2}{2m\ell^2} + mg\ell(1 - \cos\theta).$$

For an oscillation of amplitude $0 < \theta_0 < \pi$, set $k = \sin(\theta_0/2)$ and define

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2\phi}}.$$

Use energy conservation to prove

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}}\, K(k).$$

Expand the integral to obtain

$$T(\theta_0) = 2\pi\sqrt{\frac{\ell}{g}} \left(1 + \frac{\theta_0^2}{16} + O(\theta_0^4)\right).$$

Deduce the experimentally testable increase of period with amplitude and the logarithmic divergence as $\theta_0 \uparrow \pi$.

2 Spectral Theory

Definition 2.1 (Hilbert spaces and orthogonality)**.** An *inner product* on a complex linear space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies $\langle x, y\rangle = \overline{\langle y, x\rangle}$. Its norm is $\|x\| = \sqrt{\langle x, x\rangle}$. A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace $M \subseteq H$, define

$$M^\perp = \{x : \langle m, x\rangle = 0 \text{ for every } m \in M\}.$$

Definition 2.2 (Bounded operators and adjoints)**.** A linear map $T : H \to K$ between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\| \le 1} \|Tx\| < \infty.$$

Write $B(H, K)$ for the bounded operators and $B(H) = B(H, H)$. An *adjoint* of T is an operator $T^* : K \to H$ satisfying

$$\langle x, Ty\rangle = \langle T^*x, y\rangle.$$

An operator is self-adjoint when $T = T^*$, unitary when $T^*T = TT^* = I$, and normal when $T^*T = TT^*$. An *orthogonal projection* is an operator P satisfying $P = P^* = P^2$. A bounded operator is *compact* when it maps the closed unit ball to a subset with compact closure.

Exercise 2.1 (Hilbert-space foundations)**.** Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \qquad (\operatorname{im} T)^\perp = \ker T^*.$$

For every closed subspace $M \subseteq H$, prove $H = M \oplus M^\perp$ and construct the orthogonal projection $P_M : H \to M$.

Definition 2.3 (Measurable spaces)**.** A *σ-algebra* $\mathcal{F}$ on a set X contains $\varnothing$ and is closed under complements and countable unions. A *measure* is a countably additive map $\mu : \mathcal{F} \to [0, \infty]$ with $\mu(\varnothing) = 0$, and $(X, \mathcal{F}, \mu)$ is a *measure space*. A map is *measurable* when inverse images of measurable sets are measurable.

Definition 2.4 (L^p spaces)**.** For a measure space $(X, \mathcal{F}, \mu)$ and $1 \leq p < \infty$, define

$$L^p(X, \mu) = \left\{ f : \int_X |f|^p \, d\mu < \infty \right\} / \text{equality a.e.}$$

with its usual norm, and let $L^\infty(X, \mu)$ consist of essentially bounded measurable functions.

Exercise 2.2 (Lebesgue integration and convergence)**.** Construct the Lebesgue integral from nonnegative simple functions. Prove the monotone convergence theorem, Fatou's lemma, the dominated convergence theorem.

Exercise 2.3 (Product integration)**.** Construct product measure from measurable rectangles. Prove Tonelli's theorem for nonnegative functions and Fubini's theorem for integrable functions, stating the hypotheses that distinguish them. Derive the change-of-variables formula for a C^1 diffeomorphism between open subsets of finite-dimensional Euclidean spaces.

Exercise 2.4 (L^p geometry)**.** Prove Hölder's and Minkowski's inequalities and deduce that $L^p(X, \mu)$ is complete for $1 \leq p \leq \infty$. For $1 < p < \infty$ with $p^{-1} + q^{-1} = 1$, identify the continuous dual of L^p with L^q under the stated semifiniteness hypotheses.

Definition 2.5 (Gamma and beta integrals)**.** For $s, t > 0$, define

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} \, dx, \qquad B(s, t) = \int_0^1 x^{s-1} (1 - x)^{t-1} \, dx.$$

Exercise 2.5 (Beta–gamma identities)**.** Use Tonelli's theorem and the change of variables $x = ru$, $y = r(1 - u)$ to prove

$$B(s, t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s + t)}.$$

Use integration by parts to prove $\Gamma(s + 1) = s\Gamma(s)$, and use the pushforward of Euclidean measure under the norm map to obtain

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \qquad \int_{\mathbb{R}^n} e^{-|x|^2/2} \, dx = (2\pi)^{n/2}.$$

State the uses of Tonelli, Fubini, monotone convergence, and dominated convergence in the derivation.

Definition 2.6 (C^*-algebras, spectrum, and positivity)**.** A *C^*-algebra* is a complete normed complex algebra A with a conjugate-linear involution satisfying

$$(ab)^* = b^* a^*, \qquad \|ab\| \leq \|a\| \|b\|, \qquad \|a^* a\| = \|a\|^2.$$

For a unital C^*-algebra, define

$$\operatorname{spec}_A(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\}.$$

An element is *positive*, written $a \geq 0$, when $a = b^*b$ for some $b \in A$. It is *self-adjoint* when $a = a^*$ and *normal* when $a^*a = aa^*$.

Definition 2.7 (Gelfand spectrum)**.** For a commutative C^*-algebra A, let $\widehat{A}$ be the space of nonzero homomorphisms $\chi : A \to \mathbb{C}$ with its weak-$*$ topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\widehat{A}), \qquad \Gamma(a)(\chi) = \chi(a).$$

Definition 2.8 (Positive functionals and states)**.** A linear functional $\varphi : A \to \mathbb{C}$ on a unital C^*-algebra is *positive* when $\varphi(a^*a) \geq 0$ for every $a \in A$. A *state* is a positive functional satisfying $\varphi(1) = 1$.

Exercise 2.6 (Spectrum and spectral radius)**.** Prove that $\operatorname{spec}_A(a)$ is nonempty and compact and that

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}.$$

For normal a, prove $r(a) = \|a\|$.

Exercise 2.7 (Commutative Gelfand theory)**.** Prove that every character of a commutative C^*-algebra is continuous and that the Gelfand transform is an isometric $*$-isomorphism $A \cong C_0(\widehat{A})$.

Exercise 2.8 (Continuous functional calculus)**.** For every normal element a of a unital C^*-algebra, apply commutative Gelfand theory to $C^*(1, a)$ and prove that there is a unique unital isometric $*$-homomorphism

$$C(\operatorname{spec}(a)) \longrightarrow C^*(1, a), \qquad f \longmapsto f(a),$$

sending the coordinate function to a, and prove the continuous spectral mapping theorem. Deduce that positive elements have unique positive square roots and that $a \geq 0$ if and only if $a = a^*$ and $\operatorname{spec}_A(a) \subseteq [0, \infty)$.

Definition 2.9 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X is a map E from measurable subsets of X to orthogonal projections on H satisfying

$$E(\varnothing) = 0, \qquad E(X) = I, \qquad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral $\int_X f\, dE$ for bounded measurable f is defined by uniform approximation from simple functions. For a topological space X, the measure is *regular* when every scalar measure $B \mapsto \langle x, E(B)y\rangle$ is a regular Borel measure.

Exercise 2.9 (Riesz–Markov representation)**.** Let X be locally compact Hausdorff. Prove that every positive linear functional $\varphi : C_c(X) \to \mathbb{C}$ that is bounded on functions supported in each compact set has a unique regular Borel measure μ_φ satisfying

$$\varphi(f) = \int_X f\, d\mu_\varphi.$$

Use the decomposition of a bounded functional into four positive functionals to prove that every $\varphi \in C_0(X)^*$ has a unique finite complex regular measure μ_φ, and prove $\|\varphi\| = |\mu_\varphi|(X)$.

Exercise 2.10 ($C_0(X)$ representations)**.** Let X be locally compact Hausdorff. A representation $\pi : C_0(X) \to B(H)$ is *nondegenerate* when $\overline{\pi(C_0(X))H} = H$. Prove that every nondegenerate $*$-representation determines scalar measures by applying the preceding exercise to $f \mapsto \langle x, \pi(f)y\rangle$. Use polarization and multiplicativity to assemble them into a unique regular projection-valued measure E_π satisfying

$$\pi(f) = \int_X f(x)\, dE_\pi(x), \qquad f \in C_0(X).$$

Exercise 2.11 (Bounded normal spectral theorem)**.** For a normal operator $T \in B(H)$, apply the preceding representation theorem to the continuous functional calculus $C(\mathrm{spec}(T)) \to B(H)$ and prove that there is a unique projection-valued measure E_T on $\mathrm{spec}(T)$ satisfying

$$T = \int_{\mathrm{spec}(T)} z\, dE_T(z), \qquad f(T) = \int_{\mathrm{spec}(T)} f(z)\, dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space has the intrinsic spectral decomposition

$$H = \bigoplus_{\lambda \in \mathrm{spec}(T)} E_T(\{\lambda\})H, \qquad T|_{E_T(\{\lambda\})H} = \lambda I.$$

Exercise 2.12 (Compact spectral theorem)**.** For a compact normal operator K, prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_{\lambda \in \mathrm{spec}(K) \setminus \{0\}} \lambda E_K(\{\lambda\})$$

in operator norm, and prove that the spectral subspaces in the displayed sum are mutually orthogonal.

Definition 2.10 (Unbounded self-adjoint operators)**.** An unbounded operator is a linear map $A : \mathcal{D}(A) \to H$ on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax\rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For $y \in \mathcal{D}(A^*)$, the vector A^*y is determined by $\langle A^*y, x\rangle = \langle y, Ax\rangle$ for $x \in \mathcal{D}(A)$. The operator is *self-adjoint* when $A = A^*$, including equality of domains, and *closed* when its graph is closed in $H \oplus H$.

Exercise 2.13 (Cayley transform)**.** For a self-adjoint operator A, prove that $A \pm iI$ are bijective from $\mathcal{D}(A)$ to H with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with $\ker(I - C_A) = 0$. Prove conversely that every unitary C with $\ker(I - C) = 0$ determines a self-adjoint operator through the inverse Cayley transform.

Exercise 2.14 (Unbounded spectral theorem)**.** Apply the bounded normal spectral theorem to C_A and transport its projection-valued measure by the inverse Cayley map. Prove that there is a unique projection-valued measure E_A on $\mathbb{R}$ satisfying

$$A = \int_{\mathbb{R}} \lambda\, dE_A(\lambda), \qquad \mathcal{D}(A) = \left\{x : \int_{\mathbb{R}} \lambda^2\, d\langle x, E_A(\lambda)x\rangle < \infty\right\}.$$

Exercise 2.15 (Stone's theorem)**.** Use the measurable functional calculus of E_A to prove that e^{-itA} is a strongly continuous unitary group with generator $-iA$. Conversely, for a strongly continuous one-parameter unitary group, construct the spectral measure of its generator from the Fourier transforms of its scalar matrix coefficients and prove uniqueness of the self-adjoint generator.

Exercise 2.16 (Spring and string frequencies)**.** Let V be a finite-dimensional real linear space with positive-definite mass form m and nonnegative stiffness form k. Use m to define the self-adjoint operator A by $k(x, y) = m(x, Ay)$. Prove that solutions of $\ddot{x}+Ax = 0$ decompose intrinsically through the spectral projections of A and that the observed angular frequencies are $\sqrt{\lambda}$ for $\lambda \in \operatorname{spec}(A) \setminus \{0\}$.

For a stretched string of length L, wave speed c, and fixed endpoints, use the self-adjoint Dirichlet realization of $-c^2 d^2/dx^2$ to prove

$$\omega_n = \frac{n\pi c}{L}, \qquad u_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

Deduce the harmonic resonance ratios and the nodal locations measured by driving the string at each ω_n.

Exercise 2.17 (Spectral lines and time evolution)**.** Let A be self-adjoint, let $B \in B(H)$, and set $B(t) = e^{itA/\hbar} B e^{-itA/\hbar}$. For eigenvalues λ, μ of A, write $P_\lambda = E_A(\{\lambda\})$ and prove

$$P_\mu B(t) P_\lambda = e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

Use the functional calculus to prove the factorized weak identity

$$B(t) = \left(\int_{\mathbb{R}} e^{it\mu/\hbar}\, dE_A(\mu)\right) B \left(\int_{\mathbb{R}} e^{-it\lambda/\hbar}\, dE_A(\lambda)\right).$$

When A has pure-point spectrum, deduce in the weak operator topology that

$$B(t) = \sum_{\lambda,\mu} e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

For an initial state ψ, define the line strength

$$I_{\mu\lambda}(\psi) = \|P_\mu B P_\lambda \psi\|^2.$$

Prove that a line at $(\mu - \lambda)/\hbar$ occurs only when this quantity is nonzero, and identify vanishing spectral blocks as selection rules.

Definition 2.11 (Airy function)**.** The Airy function is the oscillatory integral

$$\operatorname{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt.$$

Let $a_n > 0$ be determined by $\operatorname{Ai}(-a_n) = 0$, ordered increasingly.

Exercise 2.18 (Airy spectrum of the quantum bouncer)**.** On $L^2((0, \infty), dz)$ consider the self-adjoint Dirichlet realization

$$\widehat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dz^2} + mgz, \qquad \psi(0) = 0.$$

Prove that $\operatorname{Ai}''(x) = x \operatorname{Ai}(x)$ and that Ai decays as $x \to +\infty$. With

$$\ell_g = \left(\frac{\hbar^2}{2m^2 g}\right)^{1/3},$$

derive

$$E_n = mg\ell_g a_n, \qquad \psi_n(z) = C_n \operatorname{Ai}\left(\frac{z}{\ell_g} - a_n\right).$$

Use the spectral projections of $\widehat{H}$ to prove that a periodic perturbation $B\cos(\Omega t)$ can produce a transition from the nth to the kth level only through the block $E_{\widehat{H}}(\{E_k\})BE_{\widehat{H}}(\{E_n\})$, at the resonant frequency

$$\Omega_{kn} = \frac{mg\ell_g}{\hbar}|a_k - a_n|.$$

Deduce the transmission thresholds $z_n = \ell_g a_n$ for a horizontal absorber above a gravitationally bound beam and state how measured threshold heights and resonance frequencies determine g.

3 Fourier Analysis

Definition 3.1 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Definition 3.2 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Definition 3.3 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Definition 3.4 (Locally compact groups and Haar measure)**.** A *topological group* is a group with a Hausdorff topology for which multiplication and inversion are continuous. It is *locally compact* when every point has a compact neighborhood. A *Radon measure* is a Borel measure finite on compact sets, outer regular on Borel sets, and inner regular on open sets. Every locally compact group below is equipped with a chosen nonzero left-invariant Radon measure μ, called a *left Haar measure.* Its *modular function* is determined by

$$\mu(Eg) = \Delta_G(g)^{-1}\mu(E).$$

Definition 3.5 (Regular representation)**.** Let G be a locally compact group with left Haar measure μ. For $f, g \in C_c(G)$, define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x)\, d\mu(y).$$

The *left regular representation* on $L^2(G)$ is

$$(\lambda_s\xi)(t) = \xi(s^{-1}t).$$

Exercise 3.1 (Haar regular representation)**.** Identify Haar measure on $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$. Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the L^1 convolution inequality, and $\|f * g\|_2 \leq \|f\|_1\|g\|_2$. Prove that $g \mapsto \lambda_g$ is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over G to obtain an invariant one.

Definition 3.6 (Pontryagin dual)**.** For a locally compact abelian group G, its *Pontryagin dual* $\widehat{G}$ is the group of continuous characters $\chi : G \to U(1)$, with pointwise multiplication and the compact-open topology. For $f \in L^1(G)$, define

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, d\mu(x).$$

The *double-dual map* sends $x \in G$ to the character $\chi \mapsto \chi(x)$ of $\widehat{G}$.

Exercise 3.2 (Abelian Fourier duality)**.** Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \qquad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \qquad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. Show that the group Fourier transform recovers Fourier transforms on $\mathbb{R}^n$ and Fourier series on $\mathbb{T}^n$. For a closed subgroup $H \leq G$, define its annihilator $H^\perp \subseteq \widehat{G}$ and prove $\widehat{G/H} \cong H^\perp$.

Definition 3.7 (Fourier series)**.** On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, write

$$dm_{\mathbb{T}}(x) = \frac{dx}{2\pi}$$

for normalized Haar measure and define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x)e^{-ikx}\, dm_{\mathbb{T}}(x), \qquad k \in \mathbb{Z}.$$

The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}$.

Definition 3.8 (Schwartz space and convolution)**.** The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ consists of the smooth functions f satisfying

$$\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta f(x)| < \infty$$

for all multi-indices α, β. For integrable functions, define

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)\, dy.$$

Definition 3.9 (Fourier transform)**.** Let V be an n-dimensional real Euclidean space with Lebesgue measure dq, and equip V^* with the dual measure dp. For $f \in \mathcal{S}(V)$, the semiclassical Fourier transform is

$$(\mathcal{F}_\hbar f)(p) = (2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q)\, dq.$$

Write $\mathcal{F} = \mathcal{F}_1$.

Exercise 3.3 (Fourier series and Plancherel)**.** Prove that $(e^{ikx})_{k\in\mathbb{Z}}$ is a complete orthonormal family in $L^2(\mathbb{T})$ and derive

$$f = \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}, \qquad \|f\|_2^2 = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

Exercise 3.4 (Euclidean Fourier inversion)**.** Prove Fourier inversion on $\mathcal{S}(V)$ and prove that $\mathcal{F}_\hbar$ extends uniquely to a unitary operator on $L^2(V)$. For $v \in V$, prove

$$\mathcal{F}_\hbar(-i\hbar d_v)\mathcal{F}_\hbar^{-1}\widehat{f}(p) = p(v)\widehat{f}(p).$$

Exercise 3.5 (Free-particle dispersion)**.** Substitute the plane wave $\psi(q,t) = e^{i(k(q)-\omega t)}$, with $k \in V^*$, into the free Schrödinger equation and prove

$$\hbar\omega = \frac{\hbar^2\|k\|^2}{2m}.$$

Set $p = \hbar k$ and $E = \hbar\omega$. Derive

$$E(p) = \frac{\|p\|^2}{2m}, \qquad v_{\text{group}} = \frac{p^\sharp}{m},$$

where $p^\sharp \in V$ is the Euclidean-dual vector to p. Deduce the time-of-flight displacement of a wave packet concentrated near p_0 and the de Broglie relation between its measured wavelength and momentum.

Definition 3.10 (Bessel functions)**.** For $\alpha > -1$, the Bessel function of the first kind is

$$J_\alpha(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,\Gamma(m+\alpha+1)} \left(\frac{z}{2}\right)^{2m+\alpha},$$

with a branch of z^α fixed when $\alpha \notin \mathbb{Z}$.

Exercise 3.6 (Radial Bessel transform)**.** Prove

$$z^2 J_\alpha''(z) + zJ_\alpha'(z) + (z^2 - \alpha^2)J_\alpha(z) = 0$$

and the recurrence identities

$$J_{\alpha-1}(z) + J_{\alpha+1}(z) = \frac{2\alpha}{z}J_\alpha(z), \qquad J_{\alpha-1}(z) - J_{\alpha+1}(z) = 2J_\alpha'(z).$$

Average the Euclidean Fourier kernel over spheres and express the Fourier transform of a radial Schwartz function on $\mathbb{R}^n$ as a Hankel transform with kernel $J_{n/2-1}$. Derive the corresponding inversion identity.

Exercise 3.7 (Bessel diffraction)**.** Let $j_{\nu,k}$ be the kth positive zero of J_ν. Separate the wave equation on a circular membrane of radius R with Dirichlet boundary and prove that its normal-mode frequencies are

$$\omega_{m,k} = \frac{c}{R} j_{|m|,k},$$

with radial factor $J_{|m|}(j_{|m|,k}r/R)$. Deduce the nodal circles and angular nodal lines and state how measured resonance ratios determine the ratios of Bessel zeros without knowledge of c/R.

For a uniformly illuminated circular aperture of radius a, average the Fraunhofer kernel over the aperture and derive

$$\frac{I(\theta)}{I(0)} = \left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2.$$

Prove that the dark-ring angles satisfy $ka\sin\theta_k = j_{1,k}$ and obtain the small-angle prediction $\theta_1 \sim j_{1,1}\lambda/(2\pi a)$. Identify which membrane resonances and diffraction minima measure zeros of the same Bessel functions.

Definition 3.11 (Lattices and reciprocal lattices)**.** A *lattice* in $\mathbb{R}^n$ is a subgroup $\Gamma = A\mathbb{Z}^n$ for some invertible linear map $A : \mathbb{R}^n \to \mathbb{R}^n$. Its covolume is $\mathrm{vol}(\Gamma) = |\det A|$, and a fundamental cell is $F = A[0,1)^n$. Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

Definition 3.12 (Periodic summation and structure factors)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, its Γ-*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma \in \Gamma} f(x + \gamma).$$

For a finite motif $B \subset \mathbb{R}^n$ with scattering weights c_b, its *structure factor* is

$$S_B(q) = \sum_{b \in B} c_b e^{-iq \cdot b}.$$

Exercise 3.8 (Poisson summation)**.** Prove that $P_\Gamma f$ converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\mathrm{vol}(\Gamma)} \sum_{\xi \in \Gamma^*} \mathcal{F}f(\xi) e^{i\xi \cdot x}.$$

Set $x = 0$ to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

Exercise 3.9 (Crystal diffraction and Bragg peaks)**.** Let Γ_N be a finite rectangular portion of Γ. In the single-scattering approximation, prove that the amplitude at momentum transfer q factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq \cdot \gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at $q \in \Gamma^*$ as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude $2\pi/\lambda$, derive the Laue condition and, for planes spaced by d, the Bragg relation

$$2d \sin\theta = m\lambda.$$

Show how zeros of S_B produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

Definition 3.13 (Integral lattices and theta series)**.** A *Euclidean lattice* $L \subset \mathbb{R}^d$ is a discrete subgroup for which $\mathbb{R}^d/L$ is compact. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if $L \subseteq L^*$, *unimodular* if $L = L^*$, and *even* if $\|x\|^2 \in 2\mathbb{Z}$ for every $x \in L$. For τ in the upper half-plane and $q = e^{2\pi i\tau}$, its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

Its coefficient of q^n counts lattice vectors of squared length $2n$. If the same lattice is used as a crystal lattice, its physical reciprocal lattice is

$$L^*_{\text{physical}} = 2\pi L^*.$$

Exercise 3.10 (Poisson duality for lattice theta series)**.** Apply Exercise 3.8 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\mathrm{vol}(L)} \Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives $\Theta_L(\tau+1) = \Theta_L(\tau)$. Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

Definition 3.14 (The modular group and modular forms)**.** The modular group $SL_2(\mathbb{Z})$ acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$. A *modular form of integral weight k* for $SL_2(\mathbb{Z})$ is a holomorphic function f on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion $f(\tau) = \sum_{n\geq 0} a_n q^n$ has no negative powers. It is a *cusp form* if $a_0 = 0$.

Definition 3.15 (Theta and eta functions)**.** Set

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n\in\mathbb{Z}} q^{n^2/2}, \qquad\qquad \eta(\tau) = q^{1/24}(q;q)_\infty. \\ \vartheta_4(\tau) &= \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2}, \end{aligned}$$

Exercise 3.11 (Theta transformations)**.** Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2}\vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2}\vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

Prove the T-transformation laws directly from the series and derive

$$\eta(\tau+1) = e^{\pi i/12}\eta(\tau), \qquad \eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau),$$

using the holomorphic square root on the upper half-plane. Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2/2} = \prod_{m\geq 1} (1-q^m)(1+zq^{m-1/2})(1+z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

Exercise 3.12 (Theta diffraction)**.** Let $T_L = \mathbb{R}^d/L$ carry its flat metric, and normalize L^* by $\langle x, \xi\rangle \in \mathbb{Z}$. Let $K_t(x,y)$ be the heat kernel of the flat Laplacian and prove

$$\int_{T_L} K_t(x,x)\,dx = \sum_{\xi\in L^*} e^{-4\pi^2 t\|\xi\|^2} = \Theta_{L^*}(4\pi i t).$$

Apply Poisson duality to derive the short-time image sum and its leading term $\mathrm{vol}(T_L)(4\pi t)^{-d/2}$. For a periodic array with motif B, prove that the elastic intensity at reciprocal vector $2\pi\xi$ is proportional to

$$|S_B(2\pi\xi)|^2.$$

Prove that the heat-trace exponents and diffraction-peak radii determine the same reciprocal squared lengths $\|\xi\|^2$. Deduce the predicted peak locations, systematic absences, and thermal decay rates from the theta coefficients of L^*.

4 Peter–Weyl Theory

Exercise 4.1 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 4.2 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 4.1 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \operatorname{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h\rangle_G = \frac{1}{|G|}\sum_{g\in G} \overline{f(g)}h(g).$$

Exercise 4.3 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that $\langle \chi_U, \chi_V\rangle_G = \dim \operatorname{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Definition 4.2 (Matrix coefficients)**.** A *unitary representation* of a topological group G on a Hilbert space H_π is a homomorphism $\pi : G \to U(H_\pi)$ such that $g \mapsto \pi(g)x$ is continuous for every $x \in H_\pi$. Its *matrix coefficients* are

$$g \longmapsto \langle y, \pi(g)x\rangle, \qquad x, y \in H_\pi.$$

It is *irreducible* when it has no proper nonzero closed invariant subspace. The *unitary dual* $\widehat{G}$ is the set of unitary-equivalence classes of irreducible unitary representations.

Definition 4.3 (Lie groups)**.** A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 4.4 (Exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 4.4 (Infinitesimal representations)**.** For a finite-dimensional smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \operatorname{End}(V), \qquad d\rho(X) = \left.\frac{d}{dt}\right|_0 \rho(\exp tX).$$

Exercise 4.5 (Compact convolution operators)**.** Let G be compact with normalized Haar measure and $k \in C(G)$. Prove that

$$(T_k f)(x) = \int_G k(y^{-1}x)f(y)\,dy$$

defines a compact operator on $L^2(G)$ by uniformly approximating its continuous kernel by finite sums of product functions. Prove that T_k commutes with left translations and that $k(g^{-1}) = \overline{k(g)}$ makes T_k self-adjoint. Deduce that every nonzero eigenspace of such a self-adjoint convolution operator is a finite-dimensional invariant subspace of the left regular representation.

Exercise 4.6 (Density of matrix coefficients)**.** Construct continuous convolution kernels that approximate the identity on $C(G)$. Apply the compact spectral theorem to their self-adjoint symmetrizations and prove that the linear span of matrix coefficients of finite-dimensional unitary representations is uniformly dense in $C(G)$ and dense in $L^2(G)$.

Exercise 4.7 (Schur orthogonality)**.** For irreducible finite-dimensional unitary representations π and σ, prove that

$$A \longmapsto \int_G \pi(g)A\sigma(g)^*\,dg$$

is zero when $\pi \not\simeq \sigma$ and is $A \mapsto \operatorname{tr}(A)I_{H_\pi}/\dim H_\pi$ when $\pi = \sigma$. Deduce that the coefficient map identifies $H_\pi \otimes H_\pi^*$, after multiplication by $\sqrt{\dim H_\pi}$, with an isometric subspace of $L^2(G)$, and that distinct irreducible coefficient spaces are orthogonal.

Exercise 4.8 (Peter–Weyl decomposition)**.** Use density and Schur orthogonality to prove

$$L^2(G) \cong \widehat{\bigoplus_{\pi\in\widehat{G}}} H_\pi \otimes H_\pi^*,$$

with the left regular representation acting as $\pi \otimes I_{H_\pi^*}$ on the π-summand. For $f \in L^2(G) \subseteq L^1(G)$, define $\widehat{f}(\pi) = \int_G f(g)\pi(g)^*\,dg$ and prove

$$f = \sum_{\pi\in\widehat{G}} (\dim H_\pi)\operatorname{tr}\big(\widehat{f}(\pi)\pi(\,\cdot\,)\big), \qquad \|f\|_2^2 = \sum_{\pi\in\widehat{G}} (\dim H_\pi)\operatorname{tr}\big(\widehat{f}(\pi)^*\widehat{f}(\pi)\big),$$

where the first sum converges in $L^2(G)$. Prove uniform convergence on $C(G)$ for the summability means supplied by the approximate identities in the density exercise.

Exercise 4.9 (Character inversion)**.** Let G be compact with normalized Haar measure, and let f be a finite linear combination of irreducible characters. For an irreducible unitary representation π, set

$$\widehat{f}(\pi) = \int_G f(s)\pi(s^{-1})\,ds.$$

If f is central, use Schur's lemma to write $\widehat{f}(\pi) = c_\pi I_{d_\pi}$. Derive

$$f(e) = \sum_{\pi\in\widehat{G}} d_\pi \operatorname{Tr}\big(\widehat{f}(\pi)\big) = \sum_{\pi\in\widehat{G}} d_\pi^2 c_\pi$$

and prove that convolution by f acts by the scalar c_π on every copy of π in $L^2(G)$. Extend the identity to the natural Sobolev classes for which the character series converges absolutely.

Definition 4.5 (Rotations and spin)**.** Define

$$\begin{aligned} SO(3) &= \{R \in \operatorname{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},\\ SU(2) &= \{U \in \operatorname{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}. \end{aligned}$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 4.10 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations and the 4π restoration observed by neutron interferometry.

Exercise 4.11 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \dots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \dots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$. For $j = \frac{1}{2}$, relate the two J_3 outcomes to the two Stern–Gerlach beams.

Exercise 4.12 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 4.6 (Gauss hypergeometric function)**.** For $m \in \mathbb{Z}_{\geq 0}$, define

$$(a)_0 = 1, \qquad (a)_m = a(a+1)\cdots(a+m-1) \quad (m \geq 1).$$

Where the quotient is defined, $(a)_m = \Gamma(a+m)/\Gamma(a)$. For $c \notin \mathbb{Z}_{\leq 0}$, the *Gauss hypergeometric function* is

$${}_2F_1\left(\begin{matrix} a, b \\ c \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a)_m (b)_m}{(c)_m} \frac{z^m}{m!}.$$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer and otherwise converges for $|z| < 1$. The Gauss hypergeometric operator is

$$\mathcal{L}_{a,b,c} = z(1-z)\frac{d^2}{dz^2} + \left(c - (a+b+1)z\right)\frac{d}{dz} - ab.$$

Exercise 4.13 (Gauss equation and integral)**.** Prove that ${}_2F_1(a, b; c; z)$ is the solution of $\mathcal{L}_{a,b,c} f = 0$ analytic at $z = 0$ with $f(0) = 1$. Derive Euler's integral

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\, du,$$

and Gauss's summation

$${}_2F_1(a, b; c; 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}.$$

State the parameter conditions for the integral and the summation.

Definition 4.7 (Jacobi polynomials)**.** For $\alpha, \beta > -1$, the *Jacobi polynomial* is

$$P_n^{(\alpha,\beta)}(x) = \frac{(\alpha+1)_n}{n!} {}_2F_1\left(\begin{matrix} -n, n+\alpha+\beta+1 \\ \alpha+1 \end{matrix}; \frac{1-x}{2}\right).$$

Exercise 4.14 (Jacobi spectral equation)**.** Prove

$$\left((1-x^2)\frac{d^2}{dx^2} + (\beta - \alpha - (\alpha+\beta+2)x)\frac{d}{dx}\right) P_n^{(\alpha,\beta)} = -n(n+\alpha+\beta+1) P_n^{(\alpha,\beta)}.$$

Derive Jacobi orthogonality on $[-1, 1]$ with weight $(1-x)^\alpha(1+x)^\beta$ and derive the three-term recurrence.

Exercise 4.15 (Spherical harmonics and atomic orbitals)**.** Identify S^2 with $SO(3)/SO(2)$ and $L^2(S^2)$ with the right $SO(2)$-invariant subspace of $L^2(SO(3))$. Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell$$

and prove that the functions $Y_{\ell m} \propto D^{\ell}_{m0}$, $-\ell \leq m \leq \ell$, form a complete orthogonal family in V_ℓ. For the unit round sphere, set

$$\Delta_{S^2} = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{\sin^2\theta}\partial_\phi^2.$$

Derive $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$ and completeness of the spherical harmonics. Derive the Jacobi realization

$$Y_{\ell m}(\theta,\phi) = C_{\ell m} e^{im\phi}(\sin\theta)^{|m|} P^{(|m|,|m|)}_{\ell-|m|}(\cos\theta)$$

with $C_{\ell m}$ fixed by unit norm. Separate the Coulomb Hamiltonian in spherical coordinates, identify its angular factor, and recover $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi)$. Determine the angular nodes of the s, p, and d orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare with Matsui et al., *Physical Review B* **72** (2005).

Definition 4.8 (Clebsch–Gordan and Wigner coefficients)**.** For the normalized J_3-eigenstates in the irreducible angular-momentum representations, define

$$|jm\rangle = \sum_{m_1+m_2=m} C^{jm}_{j_1m_1,j_2m_2}|j_1m_1\rangle \otimes |j_2m_2\rangle,$$

with the highest-weight coefficient positive, and set

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j+1}} C^{j,-m_3}_{j_1m_1,j_2m_2}.$$

Define the Wigner $6j$ symbol by

$$|(j_1j_2)j_{12}, j_3; jm\rangle = \sum_{j_{23}} (-1)^{j_1+j_2+j_3+j}\sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1, (j_2j_3)j_{23}; jm\rangle.$$

Exercise 4.16 (Wigner and Jacobi functions)**.** For a rotation through angle β about the second axis, set

$$d^j_{m'm}(\beta) = \langle jm'|e^{-i\beta J_2/\hbar}|jm\rangle.$$

Use the $SU(2)$ differential equation and the value at $\beta = 0$ to express $d^j_{m'm}(\beta)$ as a normalized product of powers of $\sin(\beta/2)$ and $\cos(\beta/2)$ with a Jacobi polynomial in $\cos\beta$. Deduce its orthogonality from Peter–Weyl. Use the Clebsch–Gordan decomposition to derive the linearization formula for a product of two Wigner coefficients and identify its coefficients as products of Clebsch–Gordan coefficients.

Exercise 4.17 (Angular amplitudes)**.** Derive the orthogonality relations for the Clebsch–Gordan coefficients and the triangle and magnetic selection rules for the $3j$ symbols. Prove the Gaunt identity

$$\int_{S^2} \overline{Y_{\ell_f m_f}} Y_{1q} Y_{\ell_i m_i}\, d\Omega = (-1)^{m_f}\sqrt{\frac{3(2\ell_i+1)(2\ell_f+1)}{4\pi}} \begin{pmatrix} \ell_i & 1 & \ell_f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_i & 1 & \ell_f \\ m_i & q & -m_f \end{pmatrix}.$$

Deduce the electric-dipole rules $\ell_f = \ell_i \pm 1$ and $m_f = m_i + q$, and prove that the relative Zeeman-resolved line intensities are the squared absolute values of the second $3j$ symbol after a common reduced matrix element is removed. Use the $6j$ identity to derive the corresponding relative intensities when orbital and spin angular momenta are coupled in different orders. For an oriented initial state, express the dipole radiation pattern as the squared absolute value of the resulting spherical-harmonic amplitude and identify its symmetry-enforced dark directions and absent spectral lines.

5 Harish–Chandra Theory

Definition 5.1 (Cartan and Iwasawa decompositions)**.** Let G be a connected real semisimple Lie group with finite center and let $\mathfrak{g}$ be its Lie algebra. A *Cartan involution* gives a vector space decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$$

into its $+1$ and -1 eigenspaces; the subgroup K with Lie algebra $\mathfrak{k}$ is maximal compact, and G/K is a noncompact Riemannian symmetric space. Choose a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$, put $A = \exp \mathfrak{a}$, choose positive restricted roots, and let N be the subgroup generated by their root spaces. The *Iwasawa decomposition* is the diffeomorphism

$$K \times A \times N \longrightarrow G, \qquad (k, a, n) \longmapsto kan.$$

If $\mathfrak{a}^+$ is a positive Weyl chamber, the *Cartan decomposition* is $G = K\overline{A^+}K$. Write $M = Z_K(A)$ and $W = N_K(A)/M$, and let ρ be half the sum of the positive restricted roots, counted with multiplicity.

Definition 5.2 (Spherical transform)**.** For $\lambda \in \mathfrak{a}^*$, the normalized *spherical principal series* is

$$\pi_\lambda = \mathrm{Ind}_{MAN}^{G}(1_M \otimes e^{i\lambda} \otimes 1_N),$$

where normalized induction incorporates the modular factor determined by ρ. It contains a unit K-fixed vector v_λ, and its *zonal spherical function* is

$$\varphi_\lambda(g) = \langle v_\lambda, \pi_\lambda(g) v_\lambda \rangle.$$

It is K-bi-invariant, satisfies $\varphi_\lambda(e) = 1$, and is a joint eigenfunction of the G-invariant differential operators on G/K.

For $f \in C_c^\infty(K\backslash G/K)$, its *spherical transform* is

$$\widetilde{f}(\lambda) = \int_G f(g)\varphi_{-\lambda}(g)\, dg.$$

Harish-Chandra's c-function is the coefficient governing the leading asymptotics of $\varphi_\lambda(\exp H)$ as $H \to \infty$ in a positive chamber.

Exercise 5.1 (Conical eigenfunctions on $\mathbb{H}^2$)**.** For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\mathrm{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta\varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \qquad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For a smooth compactly supported radial function, define

$$\widetilde{f}(\lambda) = 2\pi \int_0^\infty f(r)\varphi_\lambda(r)\sinh r\, dr.$$

For real λ, μ, derive the Wronskian identity

$$(\mu^2 - \lambda^2)\int_0^R \varphi_\lambda(r)\varphi_\mu(r)\sinh r\, dr = \Big[\sinh r\,(\varphi_\mu\varphi_\lambda' - \varphi_\lambda\varphi_\mu')\Big]_0^R .$$

Fix one self-adjoint boundary condition $a\varphi(R)+b\varphi'(R) = 0$, with $(a, b) \neq (0, 0)$. Prove orthogonality only for the discrete parameters λ_j satisfying this condition. Compare the large-r equation with a constant-coefficient equation and show how the discrete parameters become continuous as $R \to \infty$.

Exercise 5.2 (Mehler–Fock inversion)**.** Use the large-r asymptotics of the conical Legendre function and the Wronskian identity to derive

$$c(\lambda) = C\frac{\Gamma(i\lambda)}{\Gamma(\frac{1}{2} + i\lambda)}, \qquad |c(\lambda)|^{-2} = C'\lambda\tanh(\pi\lambda),$$

where the constants are fixed by the normalization of Haar measure. Let $C_{\mathbb{H}}$ denote the resulting inversion constant. Prove, in the distributional sense,

$$2\pi\int_0^\infty \varphi_\lambda(r)\varphi_\mu(r)\sinh r\, dr = \frac{\delta(\lambda - \mu)}{C_{\mathbb{H}}|c(\lambda)|^{-2}}.$$

Deduce the Mehler–Fock inversion and Plancherel formulas

$$f(r) = C_{\mathbb{H}}\int_0^\infty \widetilde{f}(\lambda)P_{-1/2+i\lambda}(\cosh r)|c(\lambda)|^{-2}\, d\lambda,$$

$$\|f\|_2^2 = C_{\mathbb{H}}\int_0^\infty |\widetilde{f}(\lambda)|^2|c(\lambda)|^{-2}\, d\lambda.$$

Exercise 5.3 (Hyperbolic wave propagation)**.** Let $u_{tt} = \Delta_{\mathbb{H}^2}u$ have radial initial data $u(0, r) = f(r)$ and $u_t(0, r) = g(r)$ in C_c^∞. Put $\omega_\lambda = (\lambda^2 + 1/4)^{1/2}$. Use Mehler–Fock inversion to prove

$$u(t, r) = C_{\mathbb{H}}\int_0^\infty \left(\cos(t\omega_\lambda)\widetilde{f}(\lambda) + \frac{\sin(t\omega_\lambda)}{\omega_\lambda}\widetilde{g}(\lambda)\right)P_{-1/2+i\lambda}(\cosh r)|c(\lambda)|^{-2}\, d\lambda.$$

Deduce the radial arrival profile of a localized pulse and its high-frequency spectral density from $|c(\lambda)|^{-2} = C'\lambda\tanh(\pi\lambda)$.

Exercise 5.4 (A finite hyperbolic drum)**.** Discretize the Laplace–Beltrami operator by a real symmetric weighted graph Laplacian on a finite regular hyperbolic tiling with Dirichlet boundary. Prove that the graph Hilbert space is the orthogonal direct sum of the spectral subspaces of the finite operator. Compare its measured mode ordering and pulse profile with the conical-function and wave-kernel predictions above and with the circuit measurements in Lenggenhager et al., *Nature Communications* **13** (2022). Identify discrepancies produced by the finite boundary, loss, and discretization.

6 Heisenberg Group

Definition 6.1 (Coadjoint action and stabilizers)**.** For a Lie group G with Lie algebra $\mathfrak{g}$, define

$$\mathrm{Ad}_g = d(h \mapsto ghg^{-1})_e, \qquad (\mathrm{Ad}_g^* \ell)(X) = \ell(\mathrm{Ad}_{g^{-1}} X).$$

For $\ell \in \mathfrak{g}^*$, define

$$\mathcal{O}_\ell = \mathrm{Ad}^*(G)\ell, \qquad G_\ell = \{g \in G : \mathrm{Ad}_g^* \ell = \ell\},$$

and

$$\mathfrak{g}_\ell = \{X \in \mathfrak{g} : \ell([X, Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}.$$

For $X \in \mathfrak{g}$, set

$$(\mathrm{ad}_X^* \ell)(Y) = -\ell([X, Y]).$$

Definition 6.2 (Kirillov form)**.** On $T_\ell \mathcal{O}_\ell$, define

$$\omega_\ell(\mathrm{ad}_X^* \ell, \mathrm{ad}_Y^* \ell) = \ell([X, Y]).$$

For $\eta = \mathrm{Ad}_g^* \ell$, transport ω_ℓ by the coadjoint action to define ω_η.

Exercise 6.1 (Geometry of coadjoint orbits)**.** Prove

$$T_\ell \mathcal{O}_\ell = \{\mathrm{ad}_X^* \ell : X \in \mathfrak{g}\} \cong \mathfrak{g}/\mathfrak{g}_\ell.$$

Prove that the Kirillov form is well-defined, alternating, nondegenerate, and G-invariant. Prove the cyclic identity

$$\ell([X, [Y, Z]]) + \ell([Y, [Z, X]]) + \ell([Z, [X, Y]]) = 0$$

and identify it with the closedness identity for the invariant two-form ω on $\mathcal{O}_\ell$.

Definition 6.3 (Projective representations)**.** A *normalized multiplier* on a locally compact group G is a continuous map $\sigma : G \times G \to \mathbb{T}$ satisfying

$$\sigma(x, e) = \sigma(e, x) = 1, \qquad \sigma(x, y)\sigma(xy, z) = \sigma(y, z)\sigma(x, yz).$$

A *σ-projective unitary representation* is a strongly continuous map $U : G \to U(H)$ satisfying

$$U_x U_y = \sigma(x, y) U_{xy}, \qquad U_e = I.$$

Multipliers σ and σ' are *cohomologous* when there is a continuous map $b : G \to \mathbb{T}$, $b(e) = 1$, such that

$$\sigma'(x, y) = b(x)b(y)\overline{b(xy)}\, \sigma(x, y).$$

Definition 6.4 (Weyl multipliers)**.** Let (E, ω) be a symplectic linear space. For $\lambda \in \mathbb{R}$, define

$$\sigma_{\lambda\omega}(z, z') = e^{i\lambda\omega(z,z')/2}.$$

For physical momentum variables and $\hbar > 0$, the *Weyl multiplier* is $\sigma_{\omega/\hbar}$.

Exercise 6.2 (The Weyl multiplier)**.** Prove that $\sigma_{\lambda\omega}$ is a normalized multiplier for every $\lambda \in \mathbb{R}$.

Definition 6.5 (Commutator bicharacter)**.** For a multiplier σ on an abelian group A, define

$$c_\sigma(x, y) = \sigma(x, y)\overline{\sigma(y, x)}.$$

Exercise 6.3 (Commutator bicharacter)**.** Prove that c_σ is an alternating bicharacter and depends only on the cohomology class of σ. For the Weyl multiplier on $V \oplus V^*$, prove

$$c_{\sigma_{\omega/\hbar}}\big((q,p),(q',p')\big) = e^{i(p(q')-p'(q))/\hbar}.$$

Prove

$$c_{\sigma_{\lambda\omega}}(z,z') = e^{i\lambda\omega(z,z')}.$$

Definition 6.6 (Heisenberg group)**.** For a symplectic linear space (E,ω), its *Heisenberg group* is

$$\mathbb{H}(E,\omega) = E \times \mathbb{R}$$

with multiplication

$$(z,s)(z',s') = (z+z', s+s' + \tfrac{1}{2}\omega(z,z')).$$

Write $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$ for the canonical symplectic form on $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$. Its Lie algebra is $\mathfrak{h}(E,\omega) = E \oplus \mathbb{R}Z$ with

$$[(z,sZ),(z',s'Z)] = \omega(z,z')Z.$$

For $E = V \oplus V^*$ and $\lambda \neq 0$, the *Schrödinger representation* on $L^2(V)$ is

$$\big(\pi_\lambda(q,p,s)\xi\big)(x) = e^{i\lambda(s+p(x-q/2))}\xi(x-q).$$

For $\hbar > 0$, $\ell \in V^*$, and $v \in V$, define on $\mathcal{S}(V)$

$$Q_\ell\psi(x) = \ell(x)\psi(x), \qquad P_v\psi(x) = -i\hbar\, d\psi_x(v).$$

Definition 6.7 (Polarization)**.** For a connected simply connected nilpotent group G and $\ell \in \mathfrak{g}^*$, a *polarization* is a Lie subalgebra $\mathfrak{p} \subseteq \mathfrak{g}$ satisfying

$$\ell([\mathfrak{p},\mathfrak{p}]) = 0, \qquad \dim\mathfrak{p} = \frac{\dim\mathfrak{g} + \dim\mathfrak{g}_\ell}{2}.$$

For $P = \exp(\mathfrak{p})$, set $\chi_\ell(\exp X) = e^{i\ell(X)}$. The induced representation $\operatorname{Ind}_P^G \chi_\ell$ acts by left translation on the L^2-sections over G/P satisfying $F(gp) = \chi_\ell(p)^{-1}F(g)$.

Exercise 6.4 (Central extensions)**.** Prove that $\mathbb{H}(E,\omega)$ is a locally compact group with center $\{0\} \times \mathbb{R}$ and that

$$\pi_\lambda(0,s) = e^{i\lambda s}I.$$

Prove that projective representations of E with multiplier $e^{i\lambda\omega/2}$ correspond to representations of $\mathbb{H}(E,\omega)$ with central character $e^{i\lambda s}$.

Exercise 6.5 (Canonical commutation and Weyl relations)**.** Prove

$$[Q_\ell, P_v] = i\hbar\ell(v)I.$$

For

$$(T_q\psi)(x) = \psi(x-q), \qquad (M_p^\hbar\psi)(x) = e^{ip(x)/\hbar}\psi(x),$$

prove

$$M_p^\hbar T_q = e^{ip(q)/\hbar}T_q M_p^\hbar.$$

For

$$W_\hbar(q,p) = e^{-ip(q)/(2\hbar)}M_p^\hbar T_q,$$

prove

$$W_\hbar(z)W_\hbar(z') = e^{i\omega(z,z')/(2\hbar)}W_\hbar(z+z')$$

and identify $W_\hbar(z)$ with $\pi_{1/\hbar}(z,0)$ under the canonical set-theoretic section $z \mapsto (z,0)$. Prove that this section is not a homomorphism and that its failure to preserve multiplication is exactly the displayed multiplier.

Exercise 6.6 (Schrödinger model)**.** Identify

$$\mathfrak{h}(E,\omega)^* = E^* \oplus \mathbb{R}Z^*.$$

For $(\xi,\lambda) \in E^* \oplus \mathbb{R}Z^*$, prove

$$\mathrm{Ad}^*_{(z,s)}(\xi,\lambda) = \big(\xi - \lambda\omega(z,\cdot),\lambda\big).$$

Deduce that the orbits with $\lambda = 0$ are points and that, for $\lambda \neq 0$,

$$\mathcal{O}_\lambda = \{(\xi,\lambda) : \xi \in E^*\}.$$

Prove that

$$E \longrightarrow \mathcal{O}_\lambda, \qquad z \longmapsto \big(-\lambda\omega(z,\cdot),\lambda\big),$$

pulls the Kirillov form back to $\lambda\omega$.

For $E = V \oplus V^*$ and $\mathfrak{p} = V^* \oplus \mathbb{R}Z$, prove that $\mathfrak{p}$ is a polarization at $(0,\lambda)$. Identify $\mathbb{H}(E,\omega)/\exp(\mathfrak{p})$ with V and compute the induced action to obtain

$$\big(\pi_\lambda(q,p,s)\psi\big)(x) = e^{i\lambda(s+p(x-q/2))}\psi(x-q).$$

For $\lambda = 0$, recover every character of E from a point orbit.

Exercise 6.7 (Irreducibility of the Schrödinger representation)**.** Prove that an operator on $L^2(V)$ commuting with every translation and every modulation is scalar: first use the Fourier transform to identify the commutant of translations, and then impose commutation with modulations. Deduce that π_λ is irreducible for $\lambda \neq 0$.

Exercise 6.8 (Stone–von Neumann theorem)**.** Let U be a strongly continuous irreducible unitary representation of $\mathbb{H}(V \oplus V^*,\omega)$ with central character $U(0,s) = e^{i\lambda s}I$, where $\lambda \neq 0$. Integrate Schwartz functions on $V \oplus V^*$ against the canonical section of U. Use Fourier inversion and a Gaussian approximate identity to show that this integrated representation contains the same nonzero finite-rank operators as the Schrödinger model. Construct a nonzero intertwiner and use irreducibility to prove $U \simeq \pi_\lambda$.

Definition 6.8 (Trace and Hilbert–Schmidt classes)**.** For $x, y \in H$, let

$$\theta_{x,y}(\xi) = \langle y,\xi\rangle x.$$

The *trace class* $S_1(H)$ consists of operators admitting a representation

$$A = \sum_{j=1}^\infty \theta_{x_j,y_j}, \qquad \sum_{j=1}^\infty \|x_j\|\,\|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\mathrm{Tr}(A) = \sum_{j=1}^\infty \langle y_j, x_j\rangle.$$

The *Hilbert–Schmidt class* is

$$S_2(H) = \{A \in B(H) : A^*A \in S_1(H)\}, \qquad \|A\|_2^2 = \mathrm{Tr}(A^*A).$$

Exercise 6.9 (Trace ideals)**.** Prove that $S_1(H)$, Tr, and $S_2(H)$ are well-defined. Prove that $\langle A, B\rangle_{S_2} = \mathrm{Tr}(A^*B)$ makes $S_2(H)$ a Hilbert space and that the pairing $(A, T) \mapsto \mathrm{Tr}(AT)$ identifies $S_1(H)^*$ isometrically with $B(H)$.

Definition 6.9 (Coefficient transforms)**.** Let (X, μ) be a measure space and let $U : X \to U(H)$ be a weakly measurable map. The *coefficient transform* of the unitary system U is

$$\mathcal{V}_U : S_1(H) \longrightarrow L^\infty(X), \qquad \mathcal{V}_U(A)(x) = \mathrm{Tr}(AU_x).$$

Definition 6.10 (Fourier–Wigner transform)**.** Let V be an n-dimensional real linear space with Lebesgue measure dq, let $\hbar > 0$, and equip V^* with the dual measure dp for which $(2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q)\, dq$ is unitary on L^2. For $q \in V$ and $p \in V^*$, define

$$(T_q\xi)(x) = \xi(x - q), \qquad (M_p^\hbar \xi)(x) = e^{ip(x)/\hbar}\xi(x).$$

For $z = (q, p) \in V \oplus V^*$, define

$$W_\hbar(z) = e^{-ip(q)/(2\hbar)} M_p^\hbar T_q.$$

The *Fourier–Wigner transform* is

$$\mathcal{W}_\hbar : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \qquad \mathcal{W}_\hbar(A)(z) = (2\pi\hbar)^{-n/2}\, \mathrm{Tr}(AW_\hbar(z)).$$

Exercise 6.10 (Fourier–Wigner Plancherel)**.** Prove

$$W_\hbar(z)^* = W_\hbar(-z).$$

For $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$, prove the Fourier–Wigner orthogonality identity

$$\int_{V\oplus V^*} \overline{\mathcal{W}_\hbar(A)(z)}\mathcal{W}_\hbar(B)(z)\, dz = \mathrm{Tr}(A^*B).$$

Deduce that $\mathcal{W}_\hbar$ extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever $\mathcal{W}_\hbar(A) \in L^1(V \oplus V^*)$, prove

$$A = (2\pi\hbar)^{-n/2} \int_{V\oplus V^*} \mathcal{W}_\hbar(A)(z) W_\hbar(z)^*\, dz$$

in the weak operator topology.

Exercise 6.11 (Schrödinger character)**.** For $V = \mathbb{R}^n$, let $f \in C^\infty(\mathbb{H}_n)$ satisfy

$$(q, p, s) \longmapsto f\big(\exp(q, p, s)\big) \in \mathcal{S}(\mathbb{R}^{2n+1})$$

under the canonical identification $\mathfrak{h}_n = \mathbb{R}^n \oplus (\mathbb{R}^n)^* \oplus \mathbb{R}$. With Haar measure $dq\, dp\, ds$, put

$$\pi_\lambda(f) = \int_{\mathbb{H}_n} f(g)\pi_\lambda(g)\, dg.$$

Prove

$$\mathrm{Tr}\, \pi_\lambda(f) = (2\pi)^n |\lambda|^{-n} \int_{\mathbb{R}} f(0, 0, s) e^{i\lambda s}\, ds.$$

Derive this identity directly from the Schrödinger kernel and identify the factor $|\lambda|^{-n}$ with the symplectic-volume scaling of $\mathcal{O}_\lambda$.

Exercise 6.12 (Heisenberg Plancherel and inversion)**.** For $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$, use Haar measure $dq\,dp\,ds$ and the scalar Fourier transform $\widehat{h}(\lambda) = \int_{\mathbb{R}} h(s)e^{-i\lambda s}\,ds$, and define

$$\widehat{f}(\lambda) = \int_{\mathbb{H}_n} f(g)\pi_\lambda(g)^*\,dg.$$

Compute the integral kernel of $\widehat{f}(\lambda)$ and apply Euclidean Plancherel in the central and phase-space variables. Deduce the Plancherel measure $(2\pi)^{-(n+1)}|\lambda|^n\,d\lambda$ and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R}\setminus\{0\}} \|\widehat{f}(\lambda)\|_{\mathrm{HS}}^2 |\lambda|^n\,d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

Definition 6.11 (Hermite polynomials and functions)**.** Define the physicists' Hermite polynomials by

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!}.$$

The normalized *Hermite functions* are

$$h_n(x) = \frac{e^{-x^2/2}H_n(x)}{(2^n n!\sqrt{\pi})^{1/2}}.$$

Exercise 6.13 (Mehler kernel)**.** Prove that $(h_n)_{n\geq 0}$ is a complete orthonormal family in $L^2(\mathbb{R})$ and, for $|r| < 1$, derive Mehler's identity

$$\sum_{n=0}^{\infty} r^n h_n(x)h_n(y) = \frac{1}{\sqrt{\pi(1-r^2)}}\exp\left(-\frac{(1+r^2)(x^2+y^2)-4rxy}{2(1-r^2)}\right).$$

Exercise 6.14 (Oscillator spectroscopy)**.** For

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{m\omega^2 x^2}{2},$$

use the Hermite functions to prove that

$$\psi_n(x) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right), \qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right)$$

give the complete spectral decomposition of $\widehat{H}$. Construct the creation and annihilation operators and compute

$$\langle \psi_m, Q\psi_n\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n+1}\,\delta_{m,n+1} + \sqrt{n}\,\delta_{m,n-1}).$$

Deduce the spectral-line prediction $|E_{n\pm1} - E_n|/\hbar = \omega$ and the relative line strengths for a linearly coupled oscillator. For $\alpha \in \mathbb{C}$, prove convergence and normalization of

$$\psi_\alpha = e^{-|\alpha|^2/2}\sum_{n\geq 0}\frac{\alpha^n}{\sqrt{n!}}\psi_n,$$

show that its number statistics are Poisson with mean $|\alpha|^2$, and derive its sinusoidal position expectation. Analytically continue Mehler's identity with the usual $t - i0$ prescription to obtain the oscillator propagator

$$K_t(x, y) = \sqrt{\frac{m\omega}{2\pi i\hbar \sin(\omega t)}} \exp\left[\frac{im\omega}{2\hbar \sin(\omega t)}\left((x^2 + y^2)\cos(\omega t) - 2xy\right)\right].$$

Use these formulas to predict the spectral spacing, coherent oscillation frequency, occupation-number distribution, and revival period measured in an oscillator experiment.

Part II

Deformation

7 Weyl Quantization and Wigner Tomography

Definition 7.1 (Weyl quantization)**.** For $a \in \mathcal{S}(V \oplus V^*)$, define $\mathrm{Op}_\hbar^W(a) : \mathcal{S}(V) \to \mathcal{S}(V)$ by

$$(\mathrm{Op}_\hbar^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n} \int_{V\times V^*} e^{ip(x-y)/\hbar} a\left(\frac{x+y}{2}, p\right) \psi(y)\, dy\, dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on $\mathcal{S}(V)$.

Definition 7.2 (Wigner distributions)**.** For $\psi, \varphi \in \mathcal{S}(V)$, define

$$W_\hbar(\psi, \varphi)(q, p) = \frac{1}{(2\pi\hbar)^n} \int_V e^{-ip(y)/\hbar} \overline{\psi(q - y/2)}\, \varphi(q + y/2)\, dy.$$

Write $W_\hbar[\psi] = W_\hbar(\psi, \psi)$.

Exercise 7.1 (Weyl calculus and the Moyal product)**.** For $\ell \in V^*$ and $v \in V$, prove

$$\mathrm{Op}_\hbar^W((q, p) \mapsto \ell(q)) = Q_\ell, \qquad \mathrm{Op}_\hbar^W((q, p) \mapsto p(v)) = P_v.$$

Prove

$$\mathrm{Op}_\hbar^W(a)^* = \mathrm{Op}_\hbar^W(\overline{a}), \qquad \|\mathrm{Op}_\hbar^W(a)\|_{\mathrm{HS}} = (2\pi\hbar)^{-n/2}\|a\|_2,$$

and

$$\langle \psi, \mathrm{Op}_\hbar^W(a)\varphi\rangle = \int_{V\oplus V^*} a(q, p) W_\hbar(\psi, \varphi)(q, p)\, dq\, dp.$$

Prove that for $a, b \in \mathcal{S}(V \oplus V^*)$ there is a unique $a \star_\hbar b \in \mathcal{S}(V \oplus V^*)$ satisfying

$$\mathrm{Op}_\hbar^W(a \star_\hbar b) = \mathrm{Op}_\hbar^W(a)\, \mathrm{Op}_\hbar^W(b).$$

Derive the oscillatory-integral formula for $\star_\hbar$ and prove

$$a \star_\hbar b = ab + \frac{i\hbar}{2}\{a, b\} + O(\hbar^2), \qquad \frac{a \star_\hbar b - b \star_\hbar a}{i\hbar} = \{a, b\} + O(\hbar^2)$$

in the Schwartz-symbol topology.

Exercise 7.2 (Groenewold–van Hove obstruction)**.** Specialize to $V = \mathbb{R}$ and write $Q = Q_{\mathrm{id}_{\mathbb{R}}}$ and $P = P_1$. Prove that Weyl quantization carries Poisson brackets to $(i\hbar)^{-1}$ times commutators exactly for affine and quadratic polynomials. Quantize q^2p^2 using

$$q^2p^2 = \frac{1}{9}\{q^3, p^3\} = \frac{1}{3}\{q^2p, qp^2\}$$

and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \qquad \mathcal{Q}(q) = Q, \qquad \mathcal{Q}(p) = P, \qquad \mathcal{Q}(q^m) = Q^m, \qquad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f, g\}) = (i\hbar)^{-1}[\mathcal{Q}(f), \mathcal{Q}(g)]$$

on a common invariant domain.

Definition 7.3 (Classical and quantum evolution)**.** Let $H \in C^\infty(V \oplus V^*, \mathbb{R})$ have bounded derivatives of order at least two, let Φ_t be its Hamiltonian flow, and let $\operatorname{Op}_\hbar^W(H)$ denote its self-adjoint realization. Define

$$U_\hbar(t) = e^{-it\operatorname{Op}_\hbar^W(H)/\hbar}.$$

For $a \in \mathcal{S}(V \oplus V^*)$, define

$$a_t = a \circ \Phi_t, \qquad A_\hbar(t) = U_\hbar(t)^* \operatorname{Op}_\hbar^W(a) U_\hbar(t).$$

Exercise 7.3 (Heisenberg equation and Egorov's theorem)**.** Prove

$$\frac{d}{dt}a_t = \{a_t, H\}, \qquad \frac{d}{dt}A_\hbar(t) = \frac{i}{\hbar}[\operatorname{Op}_\hbar^W(H), A_\hbar(t)].$$

For every $T > 0$, prove Egorov's theorem

$$\sup_{|t|\le T} \left\| A_\hbar(t) - \operatorname{Op}_\hbar^W(a \circ \Phi_t) \right\| = O_T(\hbar).$$

Prove exact equality for quadratic H. Derive Ehrenfest's equations for the expectations of Q_ℓ and P_v for every $\ell \in V^*$ and $v \in V$.

Definition 7.4 (Generalized Laguerre polynomials)**.** For $\alpha > -1$, define

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha}e^x}{n!}\frac{d^n}{dx^n}\left(e^{-x}x^{n+\alpha}\right), \qquad L_n = L_n^{(0)}.$$

Exercise 7.4 (Wigner–Laguerre identity)**.** Use the Laguerre Rodrigues formula to prove

$$\int_0^\infty L_m^{(\alpha)}(x) L_n^{(\alpha)}(x) x^\alpha e^{-x}\, dx = \frac{\Gamma(n+\alpha+1)}{n!}\delta_{mn}.$$

For

$$H_{\mathrm{cl}}(q,p) = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}, \qquad \psi_n(q) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{m\omega}{\hbar}}q\right),$$

prove

$$W_\hbar[\psi_n](q,p) = \frac{(-1)^n}{\pi\hbar} e^{-2H_{\mathrm{cl}}(q,p)/(\hbar\omega)} L_n\left(\frac{4H_{\mathrm{cl}}(q,p)}{\hbar\omega}\right).$$

Verify the position and momentum marginals and identify the zeros and sign changes of the Wigner distribution from those of L_n.

Exercise 7.5 (Laguerre tomography)**.** For the dimensionless quadrature

$$X_\theta = \sqrt{\frac{m\omega}{\hbar}} Q \cos\theta + \frac{P}{\sqrt{m\hbar\omega}} \sin\theta,$$

prove that its probability density in a state ψ is the Radon transform of $W_\hbar[\psi]$ along the corresponding phase-space lines. Derive the Fourier-slice inversion formula reconstructing the Wigner distribution from the family of quadrature densities. For the oscillator state ψ_n, use the Wigner–Laguerre identity to prove that the reconstructed radial profile is

$$\frac{(-1)^n}{\pi\hbar} e^{-2H_{\mathrm{cl}}/(\hbar\omega)} L_n\left(\frac{4H_{\mathrm{cl}}}{\hbar\omega}\right),$$

while every quadrature density is a scaled $e^{-x^2}H_n(x)^2$. Deduce the predicted number of quadrature nodes, the radial zero circles determined by L_n, and the sign at the phase-space origin. State how homodyne quadrature data distinguish adjacent number states and detect Wigner negativity after inverse Radon reconstruction.

Exercise 7.6 (Linear symplectic covariance)**.** Let S be a linear symplectic automorphism of $V \oplus V^*$. Construct unitary implementers for cotangent lifts of linear automorphisms of V, for symplectic shears defined by quadratic forms, and for the exchange of position and momentum implemented by the Fourier transform. Prove for these generators that

$$\mu(S)\,\mathrm{Op}_\hbar^W(a)\mu(S)^* = \mathrm{Op}_\hbar^W(a \circ S^{-1}), \qquad W_\hbar[\mu(S)\psi] = W_\hbar[\psi] \circ S^{-1}.$$

Deduce the covariance for their products and prove that each implementer is determined up to phase. Show that coherent choices form the metaplectic double cover, whose two lifts of a fixed symplectic map differ by sign. Use the Groenewold–van Hove obstruction to show why exact covariance cannot extend to every nonlinear canonical transformation. On a cotangent bundle T^*Q, apply this obstruction to overlaps of Darboux charts and identify the equivalence data required to compare their local products.

8 Moyal Deformation and Magnetic Bands

Definition 8.1 (Poisson manifolds)**.** A *Poisson manifold* is a smooth manifold M with a bracket on $C^\infty(M)$ that is a Lie bracket and a derivation in each argument. Equivalently, there is a bivector $\Pi \in \Gamma(\Lambda^2 TM)$ satisfying

$$\{f, g\} = \Pi(df, dg), \qquad [\Pi, \Pi]_{\mathrm{Sch}} = 0.$$

Definition 8.2 (Star products)**.** A *differential star product* on a Poisson manifold is an associative $\mathbb{C}[[\hbar]]$-bilinear product on $C^\infty(M)[[\hbar]]$ of the form

$$f \star g = fg + \sum_{r \geq 1} \hbar^r C_r(f, g),$$

where the C_r are bidifferential operators, 1 is a unit, and

$$C_1(f, g) - C_1(g, f) = i\{f, g\}.$$

Two star products are *equivalent* when they are intertwined by an operator $T = \mathrm{id} + \sum_{r\geq 1} \hbar^r T_r$ with differential T_r.

Exercise 8.1 (Moyal plane-wave identity)**.** On $\mathbb{R}^{2n}$ let Π be the canonical Poisson tensor and write $\Pi(k,\ell) = k_i \Pi^{ij} \ell_j$ for covectors k, ℓ. Derive from the Moyal product

$$e^{ik\cdot z} \star_\hbar e^{i\ell\cdot z} = e^{-i\hbar\Pi(k,\ell)/2} e^{i(k+\ell)\cdot z}.$$

Prove associativity directly by checking the multiplier identity for the phase. Differentiate in k and ℓ to recover the canonical commutation relations and the full bidifferential expansion of the Moyal product on polynomials.

Exercise 8.2 (Local Moyal products)**.** Cover a symplectic manifold by Darboux charts and transport the Moyal product from each chart. Prove that the local products have the same Poisson bracket on overlaps and that affine symplectic transition maps identify them exactly by linear symplectic covariance. For a general canonical transition map, compute the discrepancy through order $\hbar^2$. Formulate the recursive equations for differential operators

$$T_{ij} = \mathrm{id} + \sum_{r\geq 1} \hbar^r T_{ij,r}$$

intertwining the two local products, and derive the cocycle condition on triple overlaps required for the local products to define one global product.

Definition 8.3 (Magnetic lattice Hamiltonian)**.** Let U, V be unitary translations satisfying

$$UV = e^{2\pi i\alpha} VU,$$

where α is the magnetic flux through one lattice cell in units of the flux quantum. The magnetic nearest-neighbor Hamiltonian is

$$H_\alpha = U + U^* + V + V^*.$$

In the position realization on $\ell^2(\mathbb{Z})$, with transverse phase θ, it is the Harper operator

$$(H_{\alpha,\theta}\psi)_m = \psi_{m+1} + \psi_{m-1} + 2\cos(2\pi\alpha m + \theta)\psi_m.$$

Exercise 8.3 (Chebyshev magnetic bands)**.** Derive the magnetic-translation relation from the Moyal plane-wave identity. For rational flux $\alpha = p/q$, prove that the Harper equation has q-periodic coefficients and reduce it by the Floquet condition $\psi_{m+q} = e^{iqk}\psi_m$ to a q-dimensional spectral problem. At zero flux, use the Chebyshev identity

$$T_q(\cos t) = \cos(qt)$$

to derive the q-step discriminant

$$\Delta_{0,q}(E,\theta) = 2T_q\left(\frac{E - 2\cos\theta}{2}\right).$$

At half a flux quantum, compute the two bands

$$E_\pm(k,\theta) = \pm 2\sqrt{\cos^2 k + \cos^2\theta}.$$

Deduce the flux-dependent band edges and the touching energy, and state how microwave, photonic, or electronic lattice spectroscopy tests the Chebyshev discriminant and the half-flux splitting. Prove that the commutator phase $e^{2\pi i\alpha}$ is the phase accumulated around one lattice cell. Show that nonintegral total flux on a periodic sample obstructs consistent torus boundary phases, while integral total flux makes their corner cocycle trivial. Prove that nonzero integral flux still obstructs a single globally periodic vector potential.

9 Geometric Quantization and Landau Theta Functions

Definition 9.1 (Integration of forms)**.** An *orientation* of an n-manifold is a continuous choice of component of the nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 9.1 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form B and electric one-form E, prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt}\int_S B \qquad \text{and} \qquad dE = -\partial_t B,$$

recovering the observed flux–emf law.

Definition 9.2 (de Rham cohomology)**.** A differential form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Since $d^2 = 0$, the *de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) = \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\mathrm{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes $H^*_{\mathrm{dR}}(M)$ a graded-commutative algebra.

Exercise 9.2 (de Rham classes and periods)**.** On a star-shaped open subset of $\mathbb{R}^n$, construct a homotopy operator and prove the Poincaré lemma: every closed form of positive degree is exact. Use Stokes' theorem to prove that an exact k-form has zero integral over every closed oriented k-manifold mapped smoothly into M. Prove $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, with the isomorphism given by integration.
On $\mathbb{R}^2 \setminus \{0\}$, show that

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed and has integral 2π around the unit circle, hence is not exact. Compute $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$. Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential $F = dA$ exactly when $[F] = 0$ in de Rham cohomology.

Definition 9.3 (Connections and parallel transport)**.** A *connection* on a vector bundle $E \to M$ is an $\mathbb{R}$-bilinear map $\nabla : \Gamma(TM) \times \Gamma(E) \to \Gamma(E)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fs) = X(f)s + f\nabla_X s$. Along a curve γ it induces a covariant derivative D/dt; its solutions $Ds/dt = 0$ define *parallel transport* between endpoint fibers.

Definition 9.4 (Curvature and holonomy)**.** The *curvature* of ∇ is the $\mathrm{End}(E)$-valued two-form

$$R^\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s.$$

The *holonomy group* at p consists of the automorphisms of E_p obtained by parallel transport around loops based at p.

Exercise 9.3 (Parallel transport and curvature)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

Definition 9.5 (Principal connections)**.** For a Lie group G, a *principal G-bundle* is a smooth map $P \to M$ with a free right G-action, transitive on each fiber, and local G-equivariant trivializations. For a representation $G \to \mathrm{GL}(W)$, its associated vector bundle is $P \times_G W = (P \times W)/((pg, w) \sim (p, gw))$. A *principal connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M.

Exercise 9.4 (Gauge covariance and Bianchi identity)**.** Prove $d_A F_A = 0$. For a gauge transformation Φ, prove $F_{\Phi^* A} = \Phi^* F_A$ and that holonomy changes by conjugacy. Prove that a flat connection has trivial holonomy on contractible loops and is gauge-equivalent to a product connection on a connected base exactly when its holonomy representation is trivial. For $G = U(1)$, compute the holonomy around a generator of the fundamental group and exhibit the obstruction to removing a flat connection globally.

Definition 9.6 (Prequantum line bundles)**.** A *prequantum line bundle* over (M, ω) is a Hermitian line bundle $L \to M$ with a unitary connection ∇ satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega.$$

For $f \in C^\infty(M)$, its *Kostant–Souriau operator* is

$$\mathcal{Q}_{\mathrm{pre}}(f) = -i\hbar \nabla_{X_f} + f.$$

Definition 9.7 (Metaplectic correction)**.** A *complex polarization* is an involutive Lagrangian subbundle $P \subset T_{\mathbb{C}}M$. Its annihilator and canonical line are

$$P^\circ = \{\alpha \in T^*_{\mathbb{C}}M : \alpha|_P = 0\}, \qquad K_P = \Lambda^{\mathrm{top}} P^\circ.$$

A *half-form bundle* is a line bundle δ_P with $\delta_P^{\otimes 2} \cong K_P$, equipped with the partial connection along P induced by Lie derivatives. A corrected polarized section of $L \otimes \delta_P$ is covariantly constant along P and square-integrable for the density determined by the half-form pairing. The corrected quantum space is its Hilbert completion when this pairing is positive.

Exercise 9.5 (Prequantization and polarization)**.** Prove that a prequantum line bundle exists if and only if the period condition

$$\frac{1}{2\pi\hbar}\int_\Sigma f^*\omega \in \mathbb{Z}$$

holds for every smooth map from a closed oriented surface $f : \Sigma \to M$. Construct the line-bundle transition functions from local symplectic potentials and prove that the period condition is exactly their cocycle condition. Prove

$$[\mathcal{Q}_{\mathrm{pre}}(f), \mathcal{Q}_{\mathrm{pre}}(g)] = i\hbar \mathcal{Q}_{\mathrm{pre}}(\{f, g\}).$$

For T^*V with the vertical polarization, recover $L^2(V)$ and the operators Q_ℓ and P_v. For a compact Kähler manifold with $P = T^{0,1}M$, prove $P^\circ = T^{*1,0}M$ and $K_P = K_M$. For a positive prequantum line bundle, identify the corrected quantum space with $H^0(M, L \otimes K_M^{1/2})$.

Definition 9.8 (Theta functions with characteristics)**.** For $a, b \in \mathbb{R}$, $z \in \mathbb{C}$, and τ in the upper half-plane, define

$$\vartheta \begin{bmatrix} a \\ b \end{bmatrix} (z \mid \tau) = \sum_{n \in \mathbb{Z}} \exp\Big(\pi i (n+a)^2 \tau + 2\pi i (n+a)(z+b)\Big).$$

Exercise 9.6 (Landau theta functions)**.** Let $Q = \mathbb{R}^2/(L_x\mathbb{Z} \oplus L_y\mathbb{Z})$ have area $A = L_x L_y$, let a particle of charge e experience the constant magnetic field $B\,dx \wedge dy$, and assume $eB > 0$. Put

$$N = \frac{eBA}{2\pi\hbar}, \qquad \tau = i\frac{L_y}{L_x}, \qquad z = \frac{x}{L_x} + \tau\frac{y}{L_y}.$$

Use the prequantization condition to prove that a magnetic line bundle $L \to Q$ exists exactly when $N \in \mathbb{Z}$. For $N > 0$, use the holomorphic-gauge automorphy laws

$$f(z+1) = f(z), \qquad f(z+\tau) = e^{-\pi i N\tau - 2\pi i N z} f(z)$$

to prove the magnetic-translation decomposition

$$H^0(Q, L) = \bigoplus_{r \in \mathbb{Z}/N\mathbb{Z}} \mathbb{C}\,\vartheta \begin{bmatrix} r/N \\ 0 \end{bmatrix} (Nz \mid N\tau).$$

Derive the automorphy and magnetic-translation eigenvalue laws directly from the defining series. Relate the holomorphic-gauge sections to physical wavefunctions by their common unitary-gauge factor.

For magnetic translations by L_x/N and L_y/N, use holonomy and the Moyal plane-wave identity to prove

$$UV = e^{2\pi i/N} VU.$$

For the covariant Landau Hamiltonian, construct the raising and lowering operators and prove

$$E_n = \hbar\frac{eB}{m}\left(n + \frac{1}{2}\right), \qquad \dim E_{\widehat{H}}(\{E_n\})H = N.$$

Deduce the cyclotron resonance spacing, the degeneracy density $eB/(2\pi\hbar)$, and the magnetic-translation interference phase. State how measurements of these three quantities separately test the energy, flux-integrality, and theta-function predictions.

10 Sutherland Integrability and Jack Polynomials

Definition 10.1 (Sutherland model)**.** For N identical particles of mass m on a ring of radius R and $g > 1$, let

$$\mathrm{Conf}_N(S^1) = \{(\theta_1, \ldots, \theta_N) : \theta_i \neq \theta_j \text{ for } i \neq j\}.$$

On $C_c^\infty(\mathrm{Conf}_N(S^1))^{S_N}$ define

$$\widehat{H}_g = \frac{\hbar^2}{2mR^2}\left[-\sum_{j=1}^{N} \partial_{\theta_j}^2 + \frac{g(g-1)}{2} \sum_{i<j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}\right]$$

and let the same symbol denote its Friedrichs realization in $L^2((S^1)^N)^{S_N}$. Its ground-state factor is

$$\Psi_0(\theta) = \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^g.$$

For a nonincreasing tuple $\lambda = (\lambda_1, \dots, \lambda_N) \in \mathbb{Z}_{\geq 0}^N$, let m_λ be the sum of the distinct particle permutations of $z_1^{\lambda_1} \cdots z_N^{\lambda_N}$, where $z_j = e^{i\theta_j}$. For tuples of equal sum, write $\mu < \lambda$ in dominance order when

$$\sum_{i=1}^r \mu_i \leq \sum_{i=1}^r \lambda_i \quad (1 \leq r \leq N)$$

and at least one inequality is strict.

Definition 10.2 (Jack polynomials)**.** The Jack polynomial $P_\lambda^{(1/g)}$ is the symmetric polynomial normalized by

$$P_\lambda^{(1/g)} = m_\lambda + \sum_{\mu<\lambda} c_{\lambda\mu}(g) m_\mu$$

and by satisfying

$$\left[\sum_i (z_i\partial_{z_i})^2 + g\sum_{i<j} \frac{z_i + z_j}{z_i - z_j}(z_i\partial_{z_i} - z_j\partial_{z_j})\right] P_\lambda^{(1/g)} = \left[\sum_i (\lambda_i^2 + g(N+1-2i)\lambda_i)\right] P_\lambda^{(1/g)};$$

here $<$ is dominance order.

Exercise 10.1 (Sutherland integrability)**.** Construct a Lax equation $\dot{L} = [M, L]$ and its classical r-matrix bracket for the Sutherland system, and prove that the spectral invariants of L Poisson-commute. Conjugate the quantum Hamiltonian by Ψ_0, identify its commuting quantum integrals, and prove that

$$\Psi_\lambda = \Psi_0 P_\lambda^{(1/g)}(e^{i\theta_1}, \dots, e^{i\theta_N})$$

is a joint eigenfunction with

$$E_\lambda = \frac{\hbar^2}{2mR^2}\left[\frac{g^2 N(N^2-1)}{12} + \sum_{j=1}^N (\lambda_j^2 + g(N+1-2j)\lambda_j)\right].$$

Exercise 10.2 (Jack transition spectrum)**.** Let $\rho_r = \sum_{j=1}^N e^{ir\theta_j}$ be a density mode. Derive the Jack Pieri identity in orthonormal normalization for $\rho_1 P_\lambda^{(1/g)}$ and show that only nonincreasing tuples obtained from λ by increasing one entry by one occur. For a weak periodic drive proportional to $\rho_1 e^{-i\Omega t} + \rho_{-1}e^{i\Omega t}$, deduce the allowed resonance frequencies

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and express their relative intensities as squared normalized Jack Pieri coefficients. State how spectroscopy of particles confined to a ring determines g from the line shifts and tests the Jack selection rule from the missing lines.

11 Quantum Groups and q-Harmonic Analysis

Definition 11.1 (Poisson–Lie groups)**.** A *Poisson–Lie group* is a Lie group G with a Poisson structure for which multiplication $G \times G \to G$ is Poisson. A *Lie bialgebra* is a Lie algebra $\mathfrak{g}$ with a linear map

$$\delta : \mathfrak{g} \to \mathfrak{g} \wedge \mathfrak{g}$$

that is a 1-cocycle for the adjoint action and whose transpose defines a Lie bracket on $\mathfrak{g}^*$. It is *coboundary* when

$$\delta(x) = [x \otimes 1 + 1 \otimes x, r]$$

for some $r \in \mathfrak{g} \otimes \mathfrak{g}$.

Exercise 11.1 (Infinitesimal Poisson–Lie structure)**.** Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. For a coboundary Lie bialgebra, define

$$\mathrm{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Prove that the co-Jacobi identity is equivalent to $\mathrm{CYB}(r)$ being invariant under the diagonal adjoint action. Under the additional triangular hypothesis $\mathrm{CYB}(r) = 0$, recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as $\mathrm{CYB}(r) = \Omega$ for an invariant tensor Ω and distinguish the case $\Omega \neq 0$.

Definition 11.2 (Bialgebras and Hopf algebras)**.** A *bialgebra* is a unital associative algebra H with unital algebra homomorphisms

$$\Delta : H \to H \otimes H, \qquad \varepsilon : H \to \mathbb{C}$$

satisfying coassociativity and the counit identities. A *Hopf algebra* is a bialgebra with a linear antipode $S : H \to H$ satisfying

$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta.$$

Definition 11.3 (The compact quantum algebra $U_q(\mathfrak{sl}_2)$)**.** For $0 < q < 1$, $U_q(\mathfrak{sl}_2)$ is generated by E, F, K, K^{-1} with

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F, \qquad [E, F] = \frac{K - K^{-1}}{q - q^{-1}},$$

and

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

Its counit, antipode, and compact real-form involution are

$$\varepsilon(E) = \varepsilon(F) = 0, \quad \varepsilon(K) = 1, \quad S(E) = -K^{-1}E, \quad S(F) = -FK, \quad S(K) = K^{-1},$$

$$K^* = K, \qquad E^* = KF, \qquad F^* = EK^{-1}.$$

Exercise 11.2 (The $U_q(\mathfrak{sl}_2)$ Hopf deformation)**.** Verify that the coproduct preserves the defining relations and that the displayed counit, antipode, and involution make $U_q(\mathfrak{sl}_2)$ a Hopf $*$-algebra. Put $q = e^{-\hbar}$, expand the coproduct through first order in $\hbar$, and compute the resulting coboundary Lie bialgebra on $\mathfrak{sl}_2$. Prove that the relations and coproduct tend to those of $U(\mathfrak{sl}_2)$ as $\hbar \to 0$.

Exercise 11.3 (Quantum-group binomial identity)**.** For $YX = q^2XY$, define

$$\begin{bmatrix} n \\ r \end{bmatrix}_{q^2} = \prod_{j=1}^{r} \frac{1 - q^{2(n-r+j)}}{1 - q^{2j}}.$$

Prove the noncommutative binomial identity

$$(X + Y)^n = \sum_{r=0}^{n} \begin{bmatrix} n \\ r \end{bmatrix}_{q^2} X^r Y^{n-r}.$$

Apply it to $X = E \otimes 1$ and $Y = K \otimes E$ in $U_q(\mathfrak{sl}_2)$ to obtain an explicit formula for $\Delta(E^n)$. Show that the limit $q \to 1$ recovers the ordinary binomial theorem and the undeformed coproduct.

Definition 11.4 (Compact quantum groups and Haar states)**.** A *compact quantum group* $\mathbb{G}$ is a unital C^*-algebra $C(\mathbb{G})$ with a unital $*$-homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where $\otimes_{\min}$ is the closure of the algebraic tensor product in $B(H \otimes K)$ for faithful representations on H and K. The coproduct satisfies

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta$$

and

$$\overline{\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))} = C(\mathbb{G}) \otimes_{\min} C(\mathbb{G}) = \overline{\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)}.$$

A *Haar state* is a state h satisfying

$$(h \otimes \mathrm{id})\Delta(a) = h(a)1 = (\mathrm{id} \otimes h)\Delta(a).$$

Definition 11.5 (Representative functions)**.** A finite-dimensional *unitary corepresentation* on H_U is a unitary $U \in B(H_U) \otimes C(\mathbb{G})$ satisfying

$$(\mathrm{id} \otimes \Delta)(U) = U_{12}U_{13}.$$

For $\omega \in B(H_U)^*$, the corresponding *coefficient* is

$$u_\omega = (\omega \otimes \mathrm{id})(U) \in C(\mathbb{G}).$$

The representative-function algebra $\mathrm{Pol}(\mathbb{G})$ is the unital $*$-subalgebra generated by these coefficients for all finite-dimensional unitary corepresentations.

Definition 11.6 ($SU_q(2)$)**.** Let $\mathrm{Pol}(SU_q(2))$ be the universal unital $*$-algebra generated by α, γ so that

$$u = \begin{pmatrix} \alpha & -q\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary. Equivalently, its relations are

$$\alpha^*\alpha + \gamma^*\gamma = 1, \qquad \alpha\alpha^* + q^2\gamma\gamma^* = 1, \qquad \gamma^*\gamma = \gamma\gamma^*,$$
$$\alpha\gamma = q\gamma\alpha, \qquad \alpha\gamma^* = q\gamma^*\alpha,$$

Define $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$, and let $C(SU_q(2))$ be the universal C^*-completion.

Exercise 11.4 ($SU_q(2)$ as a compact quantum group)**.** Prove that Δ is a coassociative $*$-homomorphism satisfying the cancellation conditions. On $\ell^2(\mathbb{Z}_{\geq 0})$, define for $\zeta \in \mathbb{T}$

$$\pi_\zeta(\gamma)e_n = \zeta q^n e_n, \qquad \pi_\zeta(\alpha)e_n = (1-q^{2n})^{1/2}e_{n-1}, \qquad e_{-1} = 0.$$

Verify the relations and prove that

$$h(a) = (1-q^2)\sum_{n\geq 0} q^{2n}\int_{\mathbb{T}}\langle e_n, \pi_\zeta(a)e_n\rangle\, dm_{\mathbb{T}}(\zeta)$$

is the Haar state.

Exercise 11.5 ($SU_q(2)$ Peter–Weyl decomposition)**.** Construct the irreducible type-1 unitary $U_q(\mathfrak{sl}_2)$-modules V_n from their highest-weight lines and form their coefficient corepresentations of $\mathrm{Pol}(SU_q(2))$. Use the explicit Haar state to derive quantum Schur orthogonality and prove

$$\mathrm{Pol}(SU_q(2)) = \bigoplus_{n\geq 0} C(V_n)$$

algebraically and as a Hilbert direct sum in the Haar GNS space.

Definition 11.7 (Conjugates and quantum dimension)**.** For a finite-dimensional H-module V, its *left dual* $V^\vee$ has action

$$(x\varphi)(v) = \varphi(S(x)v),$$

evaluation $\mathrm{ev}_V(\varphi\otimes v) = \varphi(v)$, and coevaluation coev_V. In a rigid C^*-tensor category, a *conjugate* of V is an object $\overline{V}$ with morphisms

$$R_V : \mathbf{1} \longrightarrow \overline{V}\otimes V, \qquad \overline{R}_V : \mathbf{1} \longrightarrow V\otimes\overline{V}$$

satisfying

$$(\overline{R}_V^*\otimes \mathrm{id}_V)(\mathrm{id}_V\otimes R_V) = \mathrm{id}_V, \qquad (R_V^*\otimes \mathrm{id}_{\overline{V}})(\mathrm{id}_{\overline{V}}\otimes \overline{R}_V) = \mathrm{id}_{\overline{V}}.$$

The adjoints R_V^* and $\overline{R}_V^*$ are evaluation maps, and R_V and $\overline{R}_V$ are coevaluation maps. A solution is *standard* when

$$R_V^*(\mathrm{id}_{\overline{V}}\otimes T)R_V = \overline{R}_V^*(T\otimes \mathrm{id}_{\overline{V}})\overline{R}_V \qquad (T\in \mathrm{End}(V)).$$

For a standard solution, define

$$\mathrm{tr}_q(T) = R_V^*(\mathrm{id}_{\overline{V}}\otimes T)R_V, \qquad \dim_q(V) = \mathrm{tr}_q(\mathrm{id}_V).$$

Exercise 11.6 ($SU_q(2)$ fusion and dimensions)**.** Let V_n be the irreducible type-1 $U_q(\mathfrak{sl}_2)$-module of highest weight n. Construct its conjugate equations and prove

$$V_m\otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Normalize the conjugate solutions by the compact involution and compute

$$\dim_q(V_n) = [n+1]_q = \frac{q^{n+1}-q^{-(n+1)}}{q-q^{-1}}.$$

Prove

$$\dim V_n = n+1, \qquad \lim_{q\to 1}\dim_q(V_n) = n+1, \qquad \dim_q(V_n) > n+1$$

for $0<q<1$ and $n\geq 1$.

Definition 11.8 (Basic hypergeometric series)**.** For $m \in \mathbb{N}$, define

$$(a;q)_m = \prod_{j=0}^{m-1}(1-aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty}(1-aq^j),$$

$$(a_1,\dots,a_r;q)_m = \prod_{i=1}^{r}(a_i;q)_m, \qquad (a_1,\dots,a_r;q)_\infty = \prod_{i=1}^{r}(a_i;q)_\infty.$$

The *basic hypergeometric series* is

$${}_r\phi_s\left(\begin{matrix} a_1,\dots,a_r \\ b_1,\dots,b_s \end{matrix}; q, z\right) = \sum_{m=0}^{\infty} \frac{(a_1,\dots,a_r;q)_m}{(q,b_1,\dots,b_s;q)_m}\left((-1)^m q^{m(m-1)/2}\right)^{1+s-r} z^m.$$

It is terminating when one numerator parameter is q^{-n}.

Exercise 11.7 (Basic hypergeometric summations)**.** Prove the q-Gauss identity

$${}_2\phi_1\left(\begin{matrix} a,b \\ c \end{matrix}; q, \frac{c}{ab}\right) = \frac{(c/a,c/b;q)_\infty}{(c,c/(ab);q)_\infty}, \qquad \left|\frac{c}{ab}\right| < 1,$$

and the terminating q-Pfaff–Saalschütz identity

$${}_3\phi_2\left(\begin{matrix} a,b,q^{-n} \\ c,abq^{1-n}/c \end{matrix}; q, q\right) = \frac{(c/a,c/b;q)_n}{(c,c/(ab);q)_n}$$

by comparing coefficients after multiplication by the denominator q-products in the first identity, and by showing that the two sides of the second identity satisfy the same recurrence in n and have the same initial value.

Definition 11.9 (q-coupling coefficients)**.** For irreducible type-1 modules V_a, V_b, V_c, let

$$C^c_{ab} : V_c \longrightarrow V_a \otimes V_b$$

be the intertwining isometry normalized by its positive highest-weight coefficient. Its matrix coefficients between the distinguished weight lines are the *q-Clebsch–Gordan coefficients.*

Exercise 11.8 (XXZ coupling amplitudes)**.** On $V_1^{\otimes L}$ let

$$H_{\mathrm{XXZ}} = J\sum_{j=1}^{L-1}\Big(\sigma^x_j\sigma^x_{j+1} + \sigma^y_j\sigma^y_{j+1} + \frac{q+q^{-1}}{2}(\sigma^z_j\sigma^z_{j+1} - I) \\ + \frac{q-q^{-1}}{2}(\sigma^z_j - \sigma^z_{j+1})\Big).$$

For $J \in \mathbb{R}$ and $0 < q < 1$, prove that this open-chain Hamiltonian is self-adjoint and commutes with the iterated coproduct action of $U_q(\mathfrak{sl}_2)$. Decompose two coupled irreducible spin sectors by the projections $C^c_{ab}(C^c_{ab})^*$ and derive the orthogonality relations for their q-Clebsch–Gordan coefficients. Prove that their weight dependence satisfies the dual q-Hahn recurrence and identify the normalized amplitudes with a terminating ${}_3\phi_2$.

For a weak drive coupling two resolved spin sectors, derive the allowed transition strengths as squared q-Clebsch–Gordan coefficients. State the chain length, open boundary terms, resolved weights, energy scale J, and nondegeneracy assumptions under which the measured line positions and strengths determine q.

12 Braiding, Hecke Algebras, and qKZ

Definition 12.1 (Deformation parameters)**.** In this section, $0 < v < 1$ is the quantum-group deformation parameter and $\kappa \neq 0$ is the independent qKZ shift. In the next section, q is the DAHA difference parameter; in the elliptic section, p is the elliptic nome.

Definition 12.2 (Universal R-matrix and braiding)**.** A *universal R-matrix* for a Hopf algebra H is an invertible element $\mathcal{R}$ in a completion of $H \otimes H$ on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\mathrm{op}}(x), \quad (\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

For finite-dimensional modules V, W, define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}) : V \otimes W \longrightarrow W \otimes V.$$

The pair $(H, \mathcal{R})$ is *quasitriangular.*

Definition 12.3 (Spectral exchange operators)**.** Let V be finite-dimensional. A meromorphic family $R(u) \in \mathrm{End}(V \otimes V)$ satisfies the spectral Yang–Baxter equation when

$$R_{12}(u/v)R_{13}(u/w)R_{23}(v/w) = R_{23}(v/w)R_{13}(u/w)R_{12}(u/v).$$

It is regular when $R(1)$ is a nonzero scalar multiple of the flip and unitary up to scalar when $R_{12}(u)R_{21}(u^{-1})$ is scalar.

Exercise 12.1 (Hexagon identities)**.** Prove that $c_{V,W}$ is a natural module intertwiner and that the two coproduct identities for $\mathcal{R}$ give the two hexagon identities. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

and, on $V^{\otimes 3}$,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Prove that the operators $c_{j,j+1}$ define braid-group representations on tensor powers of every finite-dimensional type-1 module.

Definition 12.4 (Ribbon categories)**.** Let $\mathcal{C}$ be a braided rigid $\mathbb{C}$-linear monoidal category. A *twist* is a natural isomorphism $\theta : \mathrm{id}_{\mathcal{C}} \to \mathrm{id}_{\mathcal{C}}$ satisfying

$$\theta_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}, \qquad \theta_{X \otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object X.

Exercise 12.2 (Ribbon elements)**.** For a quasitriangular Hopf algebra $(H, \mathcal{R})$, set

$$u = m(S \otimes \mathrm{id})(\mathcal{R}_{21}).$$

Let $v \in H$ be central and invertible and satisfy

$$v^2 = uS(u), \qquad S(v) = v, \qquad \varepsilon(v) = 1, \qquad \Delta(v) = (\mathcal{R}_{21}\mathcal{R})^{-1}(v \otimes v).$$

Prove that

$$\theta_V = \pi_V(v^{-1})$$

defines a ribbon twist on the category of finite-dimensional H-modules. Derive invariance of the quantum trace under framed ribbon isotopy.

Exercise 12.3 (Chebyshev quantum dimensions)**.** For the irreducible $SU_q(2)$ objects V_n, use $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$ to prove

$$d_{n+1} = (q + q^{-1})d_n - d_{n-1}, \qquad d_0 = 1, \quad d_1 = q + q^{-1},$$

where $d_n = \dim_q(V_n)$. If U_n is the Chebyshev polynomial determined by $U_0(x) = 1$, $U_1(x) = 2x$, and $U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)$, deduce

$$\dim_q(V_n) = U_n\left(\frac{q + q^{-1}}{2}\right) = \frac{q^{n+1} - q^{-(n+1)}}{q - q^{-1}}.$$

Definition 12.5 (q-recoupling coefficients)**.** The matrix coefficients of the recoupling map

$$(V_a \otimes V_b) \otimes V_c \longrightarrow V_a \otimes (V_b \otimes V_c)$$

between fixed intermediate irreducible summands are the q-$6j$ coefficients.

Exercise 12.4 (q-Racah recoupling)**.** Compute the change between the two coupling orders of $V_a \otimes V_b \otimes V_c$ and express its entries as terminating ${}_4\phi_3$ series with base determined by v. Identify them with q-Racah polynomials and derive their orthogonality from unitarity of the recoupling map for the compact real form. Prove the pentagon identity by comparing the five coupling orders of four factors. For three resolved XXZ spin sectors, derive the recoupling probabilities as squared q-$6j$ coefficients and state the unitarity and spectral-resolution assumptions required for this probability interpretation.

Definition 12.6 (Type-A Hecke actions)**.** For $v \neq 0$, the Hecke algebra $\mathcal{H}_v(S_r)$ is generated by $T_1, \ldots, T_{r-1}$ with

$$T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \qquad T_iT_j = T_jT_i \quad (|i-j| > 1),$$
$$(T_i - v)(T_i + v^{-1}) = 0.$$

Let V_1 be the fundamental $U_v(\mathfrak{sl}_2)$-module. On $V_1 \otimes V_1 \cong V_2 \oplus V_0$, let P_2, P_0 be the invariant orthogonal projections, and define

$$T = vP_2 - v^{-1}P_0, \qquad \rho_r(T_i) = \mathrm{id}^{\otimes(i-1)} \otimes T \otimes \mathrm{id}^{\otimes(r-i-1)}.$$

Exercise 12.5 (Fundamental Hecke symmetry)**.** Prove that the coproduct action of $U_v(\mathfrak{sl}_2)$ commutes with $\rho_r(T_i)$. Derive the braid relation from the q-Racah recoupling identity and the Hecke relation from the two displayed spectral projections. Put $e_i = (v - T_i)/(v + v^{-1})$ and prove that the action factors through the Temperley–Lieb relations

$$e_i^2 = e_i, \qquad e_ie_{i\pm1}e_i = (v + v^{-1})^{-2}e_i.$$

Definition 12.7 (Affine Hecke algebras)**.** The extended affine Hecke algebra $\mathcal{H}_v^{\mathrm{aff}}(GL_r)$ is generated by $T_1, \ldots, T_{r-1}$ and commuting invertible elements $X_1, \ldots, X_r$, subject to the finite Hecke relations and

$$T_iX_iT_i = X_{i+1}, \qquad T_iX_j = X_jT_i \quad (j \neq i, i+1).$$

Equivalently, it is the Hecke quotient of the extended affine braid group of $\mathbb{Z}^r \rtimes S_r$.

Definition 12.8 (Baxterized Hecke operators)**.** For an invertible spectral parameter u, define

$$\check{R}_i(u) = uT_i - u^{-1}T_i^{-1}.$$

Its eigenvalues on the v and $-v^{-1}$ eigenspaces of T_i are

$$uv - u^{-1}v^{-1}, \qquad -uv^{-1} + u^{-1}v.$$

Exercise 12.6 (Baxterization)**.** Use the braid and Hecke relations for T_i to prove

$$\check{R}_i(u)\check{R}_{i+1}(uv)\check{R}_i(v) = \check{R}_{i+1}(v)\check{R}_i(uv)\check{R}_{i+1}(u).$$

Prove that $\check{R}_i(1)$ is a scalar multiple of the identity and compute $\check{R}_i(u)\check{R}_i(u^{-1})$. For $|u| = 1$, prove that $\check{R}_i(u)^* = \check{R}_i(u^{-1})$ and normalize the operators to be unitary wherever the scalar product is nonzero. Prove that their two eigenvalues depend on u and hence that the spectral family itself does not satisfy one parameter-independent Hecke polynomial.

Definition 12.9 (qKZ transport)**.** Let v be the quantum-group parameter and let $\kappa \neq 0$ be an independent shift parameter. Let $D \in \mathrm{End}(V)$ be invertible and satisfy $[D \otimes D, \check{R}(u)] = 0$. Let D_1 act on the transported tensor factor and define

$$\mathcal{K}_1(z_1, \ldots, z_r) = D_1\check{R}_{1r}(z_1/z_r) \cdots \check{R}_{12}(z_1/z_2).$$

The first qKZ equation is

$$\Psi(z_2, \ldots, z_r, \kappa z_1) = \mathcal{K}_1(z_1, \ldots, z_r)\Psi(z_1, \ldots, z_r),$$

and the remaining equations are obtained by cyclic conjugation.

Exercise 12.7 (qKZ exchange transport)**.** Use Baxterization to prove consistency of the adjacent exchange rules and the qKZ difference equations. Identify the constant exchanges and winding operators with the generators of $\mathcal{H}_v^{\mathrm{aff}}(GL_r)$, keeping v and κ independent. For the unitary normalization, prepare and detect three resolved XXZ spin sectors and prove that the two factorizations of a three-spin exchange give the same output probabilities. Express this equality as the q-Racah recoupling identity. Derive the phase accumulated in one κ-shift cycle and state the twist, unitarity, preparation, and detection assumptions under which it is an interference observable.

13 DAHA and Macdonald–Ruijsenaars Theory

Definition 13.1 (Double-affine braid group)**.** Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ and let

$$\mathrm{Conf}_r(T^2) = \{(z_1, \ldots, z_r) \in (T^2)^r : z_i \neq z_j \text{ for } i \neq j\}.$$

The *elliptic braid group* is

$$B_r^{\mathrm{ell}} = \pi_1(\mathrm{Conf}_r(T^2)/S_r).$$

Its standard generators are adjacent braids T_i and loops X_j, Y_j moving the jth point around the two fundamental cycles of T^2. The central extension in which the total X–Y commutator is denoted q is the *double-affine braid group* of type GL_r.

Exercise 13.1 (From affine to double-affine braids)**.** Derive a presentation of B_r^{ell} from the motions T_i, X_j, Y_j. Prove that the T_i satisfy the finite braid relations, that the X_j commute, that the Y_j commute, and that conjugation by T_i permutes the adjacent loop generators. Compute the central commutator of the two total torus motions. Cutting T^2 along one cycle, prove that the remaining generators form an affine braid group.

Definition 13.2 (Type-A double affine Hecke algebra)**.** For nonzero parameters q, t, the *double affine Hecke algebra* $\mathsf{H}_{q,t}(GL_r)$ is the quotient of the central extension of $\mathbb{C}(q,t)[B_r^{\mathrm{ell}}]$ in the preceding definition by

$$\mathsf{q} = q, \qquad (T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

Definition 13.3 (Rank-one DAHA)**.** The universal DAHA $\widehat{\mathsf{H}}_q(C_1^\vee, C_1)$ is generated by $t_i^{\pm 1}$ for $0 \le i \le 3$, subject to

$$t_i t_i^{-1} = 1, \qquad t_i + t_i^{-1} \text{ central}, \qquad t_0 t_1 t_2 t_3 = q^{-1}.$$

For $k_i \neq 0$, the specialization

$$t_i + t_i^{-1} = k_i + k_i^{-1}$$

is the four-parameter rank-one DAHA. After the standard scalar normalization, let T be its finite Hecke generator with $(T-k)(T+k^{-1}) = 0$, and define

$$e = \frac{T + k^{-1}}{k + k^{-1}}.$$

Its basic polynomial representation is on $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$ by reflection–q-difference operators, and

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}].$$

The action of $e\widehat{\mathsf{H}}_q e$ on $e\mathcal{L}$ is the *spherical rank-one polynomial representation.*

Definition 13.4 (Generalized hypergeometric series)**.** If no denominator parameter b_j is a nonpositive integer, define

$${}_rF_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_r)_m}{(b_1)_m \cdots (b_s)_m} \frac{z^m}{m!}.$$

The series is finite when a numerator parameter is a nonpositive integer. Otherwise it converges for every z when $r \le s$, for $|z| < 1$ when $r = s+1$, and only at $z = 0$ when $r > s+1$.

Definition 13.5 (Askey–Wilson operator and polynomials)**.** For $0 < q < 1$ and nonzero a, b, c, d, define

$$A(z) = \frac{(1-az)(1-bz)(1-cz)(1-dz)}{(1-z^2)(1-qz^2)}$$

and, on $f(z) = f(z^{-1})$,

$$(\mathcal{D}_{AW} f)(z) = A(z)\big(f(qz) - f(z)\big) + A(z^{-1})\big(f(q^{-1}z) - f(z)\big).$$

For $x = (z + z^{-1})/2$, the *Askey–Wilson polynomial* is

$$\begin{aligned} p_n(x; a, b, c, d \mid q) = {} & a^{-n}(ab, ac, ad; q)_n \\ & \times {}_4\phi_3\left(\begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix}; q, q\right). \end{aligned}$$

Its weight on the unit circle is

$$w_{AW}(z) = \frac{(z^2, z^{-2}; q)_\infty}{(az, a/z, bz, b/z, cz, c/z, dz, d/z; q)_\infty}.$$

Exercise 13.2 (Askey–Wilson theorem)**.** Construct the basic representation of $\widehat{\mathsf{H}}_q(C_1^\vee, C_1)$ from the involutions

$$(s_1 f)(z) = f(z^{-1}), \qquad (s_0 f)(z) = f(qz^{-1})$$

and first-order reflection–q-difference operators. Verify the four Hecke relations and the product relation. Prove that the spherical subalgebra is generated in its polynomial representation by multiplication by $z + z^{-1}$ and $\mathcal{D}_{AW}$. Prove

$$\mathcal{D}_{AW} p_n = (q^{-n} - 1)(1 - abcdq^{n-1}) p_n.$$

For $a, b, c, d \in (-1, 1)$, prove that $\mathcal{D}_{AW}$ is symmetric for

$$\langle f, g \rangle_{AW} = \int_{|z|=1} f(z)\overline{g(z)} w_{AW}(z) \frac{dz}{2\pi i z},$$

and deduce the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

Definition 13.6 (Wilson polynomials)**.** For positive a, b, c, d, define

$$\begin{aligned} W_n(x^2; a, b, c, d) &= (a + b)_n (a + c)_n (a + d)_n \\ &\quad \times {}_4F_3\left(\begin{matrix} -n, n + a + b + c + d - 1, a + ix, a - ix \\ a + b, a + c, a + d \end{matrix}; 1 \right). \end{aligned}$$

Exercise 13.3 (Wilson degeneration)**.** Set $q = e^{-\varepsilon}$ and scale the Askey–Wilson parameters and spectral variable by

$$(a, b, c, d, z) = (q^A, q^B, q^C, q^D, q^{ix}).$$

After the scalar normalization by $(1 - q)^{-3n}$, prove that the Askey–Wilson polynomial tends to $W_n(x^2; A, B, C, D)$ as $\varepsilon \downarrow 0$. Derive the Wilson difference equation, weight, orthogonality, and three-term recurrence as limits of the corresponding Askey–Wilson identities.

Definition 13.7 (Type-A polynomial representation)**.** Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \ldots, x_r^{\pm 1}],$$

let X_j act by multiplication by x_j, and let s_i exchange x_i and x_{i+1}. Define

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i / x_{i+1} - 1}, \qquad 1 \le i < r,$$

and

$$(\pi f)(x_1, \ldots, x_r) = f(x_2, \ldots, x_r, q x_1).$$

For generic independent parameters q, t, the operator algebra generated by $X_j^{\pm 1}$, T_i, and $\pi^{\pm 1}$ is the polynomial realization of the GL_r *double affine Hecke algebra* $\mathsf{H}_{q,t}(GL_r)$. The subalgebra generated by T_i and $\pi^{\pm 1}$ is an extended affine Hecke algebra; its translation elements form a commuting family $Y_1, \ldots, Y_r$ of q-difference-reflection operators.

Exercise 13.4 (DAHA polynomial quotient)**.** Prove that the displayed operators preserve $\mathcal{P}_r$ and satisfy the braid and Hecke relations. Prove

$$\pi X_i \pi^{-1} = X_{i+1} \quad (1 \le i < r), \qquad \pi X_r \pi^{-1} = q X_1,$$

and identify the two affine-Hecke subalgebras generated respectively by T_i, X_j and by T_i, Y_j. Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to q.

Definition 13.8 (Macdonald operators and polynomials)**.** Let

$$(T_{q,x_i}f)(x_1,\dots,x_r) = f(x_1,\dots,qx_i,\dots,x_r).$$

The first type-A *Macdonald operator* is

$$D_{q,t} = \sum_{i=1}^{r}\left(\prod_{j\neq i}\frac{tx_i - x_j}{x_i - x_j}\right)T_{q,x_i}.$$

For generic q,t, the *Macdonald polynomial* $P_\lambda(x;q,t)$ is the symmetric polynomial triangular in the monomial symmetric functions and satisfying

$$D_{q,t}P_\lambda = \left(\sum_{i=1}^{r} q^{\lambda_i}t^{r-i}\right)P_\lambda, \qquad \lambda_1 \geq \cdots \geq \lambda_r \geq 0.$$

For $0<q,t<1$, define

$$\Delta_{q,t}(x) = \prod_{i\neq j}\frac{(x_i/x_j;q)_\infty}{(tx_i/x_j;q)_\infty},$$

and

$$\langle f,g\rangle_{q,t} = \frac{1}{r!}\int_{(S^1)^r} f(x)\overline{g(x)}\Delta_{q,t}(x)\prod_{j=1}^{r}\frac{dx_j}{2\pi i x_j}.$$

Exercise 13.5 (Cherednik operators)**.** Define

$$Y_i = T_i\cdots T_{r-1}\pi T_1^{-1}\cdots T_{i-1}^{-1}.$$

Use the displayed polynomial representation and the affine braid relations to prove that the Y_i commute. Prove that their symmetric Laurent polynomials preserve $\mathcal{P}_r^{S_r}$ and identify $t^{-(r-1)/2}(Y_1 + \cdots + Y_r)$ with $D_{q,t}$.

Exercise 13.6 (Macdonald spectral theory)**.** Order monomial symmetric functions by dominance and prove that $D_{q,t}$ is triangular with the displayed diagonal entries. For generic q,t, deduce existence and uniqueness of P_λ. Prove directly from the torus integral that $D_{q,t}$ is symmetric when $0<q,t<1$, and deduce orthogonality and joint diagonalization for symmetric Laurent polynomials in the commuting Y_i.

Definition 13.9 (Multiplicative theta function)**.** For $|p|<1$, define

$$\theta(z;p) = (z;p)_\infty(p/z;p)_\infty.$$

Definition 13.10 (Elliptic Ruijsenaars operator)**.** For type A_{r-1}, define

$$D^{\mathrm{ell}}_{p,q,t} = \sum_{i=1}^{r}\left(\prod_{j\neq i}\frac{\theta(tx_i/x_j;p)}{\theta(x_i/x_j;p)}\right)T_{q,x_i}.$$

Exercise 13.7 (Ruijsenaars degeneration)**.** Prove directly from $\theta(z;0)=1-z$ that

$$D^{\mathrm{ell}}_{p,q,t} \longrightarrow D_{q,t} \qquad (p\to 0).$$

Set $q=e^{\varepsilon}$ and $t=e^{g\varepsilon}$ and derive the Heckman–Opdam differential operator from the second-order term as $\varepsilon\to 0$. Take the rational scaling limit of its hyperbolic coefficients and obtain the rational Calogero–Moser operator.

Exercise 13.8 (The Calogero–Sutherland limit)**.** Write $x_i = e^{z_i}$, set $q = e^{\epsilon}$ and $t = e^{g\epsilon}$, and expand $D_{q,t}$ through order ϵ^2. After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i<j} \coth\left(\frac{z_i - z_j}{2}\right)(\partial_{z_i} - \partial_{z_j}).$$

On the circle, put $z_i = i\theta_i$ and conjugate by

$$\Psi_0(\theta) = \prod_{i<j} \left| 2\sin\frac{\theta_i - \theta_j}{2} \right|^g.$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = -\sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i<j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for $r = 2$, degree two. Identify their Sutherland eigenvalues.

Exercise 13.9 (Hypergeometric degeneration)**.** Set $q = e^{-\varepsilon}$. Prove

$$\lim_{\varepsilon \to 0^+} \frac{(q^a;q)_m}{(1-q)^m} = (a)_m, \qquad \lim_{q \to 1^-} (1-q)^{1-a} \frac{(q;q)_\infty}{(q^a;q)_\infty} = \Gamma(a).$$

Deduce

$$\lim_{q \to 1^-} {}_r\phi_{r-1}\left(\begin{matrix} q^{a_1}, \dots, q^{a_r} \\ q^{b_1}, \dots, q^{b_{r-1}} \end{matrix} ; q, z \right) = {}_rF_{r-1}\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix} ; z \right).$$

Derive Gauss's summation from the q-Gauss summation. Prove the corresponding degeneration of the basic hypergeometric q-difference equation to the Gauss equation.

Definition 13.11 (BC_r root-system weight)**.** Write $f(z^{\pm m}) = f(z^m)f(z^{-m})$ and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon,\delta = \pm 1} f(z_i^{\epsilon} z_j^{\delta}).$$

For $|q|, |t|, |a_s| < 1$, define

$$\Delta_{BC_r}(z) = \prod_{i=1}^r \frac{(z_i^{\pm 2};q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1};q)_\infty} \times \prod_{1 \le i < j \le r} \frac{(z_i^{\pm 1} z_j^{\pm 1};q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1};q)_\infty}.$$

Use the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

Exercise 13.10 (BC_r weights and limits)**.** Prove that Δ_{BC_r} is invariant under permutations and independent inversions of the z_i. Identify its factors with the roots $\pm e_i$, $\pm 2e_i$, and $\pm e_i \pm e_j$. For $r = 1$, recover the Askey–Wilson weight and its order-two Weyl normalization. Set $t = q^g$ with $g \in \mathbb{N}$, reduce the quotients of q-products to finite products, and compute the limit $q \to 1^-$ with $z_i = e^{i\theta_i}$.

Exercise 13.11 (Askey–Wilson spectrum)**.** Fix $M \geq 0$ and let $\mathcal{H}_M$ be the span of the normalized Askey–Wilson polynomials $p_0, \ldots, p_M$ with their positive Askey–Wilson inner product. Model a finite programmable q-difference simulator by the self-adjoint Hamiltonian $\widehat{H}_M = \varepsilon \mathcal{D}_{AW}|_{\mathcal{H}_M}$, where $\varepsilon > 0$ is a calibrated energy scale. Prove that its spectral values are

$$E_n = \varepsilon(q^{-n} - 1)(1 - abcdq^{n-1}).$$

Write the three-term recurrence in the form

$$(z + z^{-1})p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

and compute the corresponding normalized matrix elements. Deduce that a periodic perturbation proportional to the compression of $z + z^{-1}$ to $\mathcal{H}_M$ produces only neighboring transitions, with resonance frequencies $|E_{n+1} - E_n|/\hbar$ and intensities determined by A_n and C_{n+1}. Assuming ε is known and sufficiently many nondegenerate lines are resolved, derive the equations satisfied by q and the product $abcd$. Compute the Jacobian of two independent gap ratios and state the nonvanishing and parameter-range hypotheses under which these two quantities are locally identifiable. Prove that the relative strengths determine only the unordered parameter set $\{a, b, c, d\}$ unless additional observables break its permutation symmetry. State that these are predictions for the engineered simulator.

Exercise 13.12 (Macdonald–Ruijsenaars spectroscopy)**.** Let $0 < q, t < 1$, fix $M \geq 0$, and define

$$\mathcal{H}_{q,t}^{(M)} = \operatorname{span}\{P_\lambda(x; q, t) : |\lambda| \leq M\}$$

with the positive Macdonald inner product. Put

$$d_0 = \sum_{i=1}^{r} t^{r-i}, \qquad \widehat{H}_{q,t}^{(M)} = \varepsilon(d_0 - D_{q,t})|_{\mathcal{H}_{q,t}^{(M)}}, \qquad \varepsilon > 0.$$

Identify $D_{q,t}$ with the first symmetric function of the commuting DAHA winding operators and derive its compatibility with the Hecke exchange relations. Prove that $\mathcal{H}_{q,t}^{(M)}$ is invariant, that $\widehat{H}_{q,t}^{(M)}$ is self-adjoint, and that the normalized Macdonald states have energies

$$E_\lambda = \varepsilon \sum_{i=1}^{r} t^{r-i}(1 - q^{\lambda_i}).$$

Model a finite programmable q-difference simulator by this operator. Use the Macdonald Pieri identity for $\rho_1 = x_1 + \cdots + x_r$ and its adjoint to determine every nonzero matrix element of the compression of $\rho_1 + \rho_1^*$ to $\mathcal{H}_{q,t}^{(M)}$. For a periodic drive proportional to this compressed observable, derive the resonance frequencies

$$\Omega_{\mu\lambda} = |E_\mu - E_\lambda|/\hbar$$

and the relative intensities as squared normalized Macdonald Pieri coefficients. Set $q = e^{-\delta}$ and $t = e^{-g\delta}$; after the scalar and center-of-mass normalizations in the preceding exercise, prove that the rescaled spectral gaps and transition coefficients converge to the Sutherland energies and Jack Pieri coefficients. State how measured line locations and strengths test this degeneration. Explain that recovery of q, t requires known ε and r, resolved partition labels, sufficiently many nondegenerate transitions, and a fixed convention for any discrete parameter ambiguity, and that the prediction concerns the engineered simulator.

14 Elliptic Hypergeometric Integrability

Definition 14.1 (Elliptic shifted factorials and gamma function)**.** For $|p|, |q| < 1$, define

$$\theta(z_1, \ldots, z_s; p) = \prod_{j=1}^{s} \theta(z_j; p),$$

$$(a; q, p)_m = \prod_{j=0}^{m-1} \theta(aq^j; p),$$

$$(a_1, \ldots, a_s; q, p)_m = \prod_{i=1}^{s} (a_i; q, p)_m,$$

and

$$\Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1} q^{k+1}/z}{1 - p^j q^k z}.$$

Definition 14.2 (Deformation hierarchy)**.** For a series $\sum_m c_m$, set $h_m = c_{m+1}/c_m$. It is *ordinary hypergeometric* when h_m is rational in m, and *basic hypergeometric* when h_m is rational in q^m. It is *elliptic hypergeometric* when, after writing $x = q^m$, its term ratio $h(x)$ is multiplicatively p-periodic:

$$h(px) = h(x).$$

Its parameters are *balanced* when the products of the numerator and denominator parameters required by this ellipticity agree. The deformation and degeneration orders are

$$\text{ordinary/additive} \longrightarrow \text{basic/multiplicative} \longrightarrow \text{elliptic},$$

$$\text{elliptic} \xrightarrow{p \to 0} \text{basic} \xrightarrow{q \to 1} \text{ordinary}.$$

Exercise 14.1 (Elliptic degenerations)**.** Prove

$$\theta(pz; p) = -z^{-1} \theta(z; p), \qquad \Gamma(qz; p, q) = \theta(z; p) \Gamma(z; p, q),$$

$$\Gamma(pz; p, q) = \theta(z; q) \Gamma(z; p, q), \qquad \Gamma(z; p, q) \Gamma(pq/z; p, q) = 1.$$

Prove

$$\theta(z; 0) = 1 - z, \qquad (a; q, 0)_m = (a; q)_m, \qquad \Gamma(z; 0, q) = \frac{1}{(z; q)_\infty}.$$

Definition 14.3 (Frenkel–Turaev sum)**.** For $a^2 q^{n+1} = bcde$, define

$$\begin{aligned}{}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) = \sum_{m=0}^{n} & \frac{\theta(aq^{2m}; p)}{\theta(a; p)} \\ & \times \frac{(a, b, c, d, e, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq^{n+1}; q, p)_m} q^m.\end{aligned}$$

Exercise 14.2 (Theta addition and Frenkel–Turaev summation)**.** Prove the multiplicative theta addition formula

$$\begin{aligned}&\theta(xz, x/z, yw, y/w; p) - \theta(xw, x/w, yz, y/z; p) \\ &= \frac{y}{z} \theta(xy, x/y, zw, z/w; p)\end{aligned}$$

by comparing quasi-periodicity and zeros. Use it to telescope the difference between the nth and $(n-1)$st terminating sums and prove

$$ {}_{10}V_9(a;b,c,d,e,q^{-n};q,p) = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q,p)_n}{(aq/b, aq/c, aq/d, aq/(bcd); q,p)_n}. $$

Take $p \to 0$ and identify the terminating very-well-poised basic hypergeometric summation.

Definition 14.4 (Terminating elliptic weights)**.** Assume $a^2 q^{n+1} = bcde$ and $0 \le m \le n$. Define

$$ S_e(m) = \frac{\theta(aq^{2m};p)}{\theta(a;p)} \frac{(a, e, q^{-n}; q,p)_m}{(q, aq/e, aq^{n+1}; q,p)_m} q^m, $$

$$ W_b(m) = \frac{(b;q,p)_m}{(aq/b;q,p)_m}, \quad W_c(m) = \frac{(c;q,p)_m}{(aq/c;q,p)_m}, \quad W_d(m) = \frac{(d;q,p)_m}{(aq/d;q,p)_m}, $$

and the triangle weight

$$ \mathcal{T}_n(a;b,c,d) = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q,p)_n}{(aq/b, aq/c, aq/d, aq/(bcd); q,p)_n}. $$

Exercise 14.3 (Terminating elliptic contraction)**.** Rewrite the Frenkel–Turaev summation as the explicitly normalized finite contraction

$$ \sum_{m=0}^{n} S_e(m) W_b(m) W_c(m) W_d(m) = \mathcal{T}_n(a;b,c,d). $$

Regard the summands as the weights of $n+1$ internal states in a programmable finite statistical model. For positive specialized weights, derive the normalized occupation probabilities and prove that the measured partition function is exactly $\mathcal{T}_n(a;b,c,d)$. Take $p \to 0$ and then $q \to 1$ and compute the corresponding multiplicative and additive limits. Determine how zeros of the shifted factorials truncate the internal-state sum when q approaches a root of unity.

15 Root-of-Unity Specialization and Verlinde Counting

Definition 15.1 (Temperley–Lieb algebras)**.** For $d \in \mathbb{C}$, the Temperley–Lieb algebra $TL_n(d)$ is generated by $U_1, \dots, U_{n-1}$ with

$$ U_i^2 = dU_i, \qquad U_i U_{i\pm1} U_i = U_i, \qquad U_i U_j = U_j U_i \quad (|i-j| > 1). $$

Its diagram trace is normalized by

$$ \mathrm{tr}_0(1) = 1, \qquad \mathrm{tr}_{n+1}(x) = d\,\mathrm{tr}_n(x), \qquad \mathrm{tr}_{n+1}(xU_n) = \mathrm{tr}_n(x), $$

where $x \in TL_n(d)$ is embedded in $TL_{n+1}(d)$.

Definition 15.2 (Jones–Wenzl recursion)**.** Put

$$ q = e^{\pi i/(k+2)}, \qquad d = q + q^{-1}, \qquad [n]_q = \frac{q^n - q^{-n}}{q - q^{-1}}. $$

With $f_0 = f_1 = 1$, define recursively in $TL_{n+1}(d)$

$$ f_{n+1} = f_n - \frac{[n]_q}{[n+1]_q} f_n U_n f_n \qquad (1 \le n \le k). $$

Let $\mathcal{C}_k$ be the idempotent-complete tensor quotient of the Temperley–Lieb category by the tensor ideal generated by f_{k+1}, and write V_a for the image of f_a, $0 \le a \le k$.

Exercise 15.1 (Jones–Wenzl truncation)**.** Use the Temperley–Lieb relations to prove inductively

$$f_n^2 = f_n, \qquad U_i f_n = f_n U_i = 0 \quad (1 \le i < n), \qquad \mathrm{tr}(f_n) = [n+1]_q.$$

Deduce $\mathrm{tr}(f_{k+1}) = [k+2]_q = 0$ and prove that imposing $f_{k+1} = 0$ makes the recursion close on $V_0, \dots, V_k$. Derive

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b| \le c \le \min(a+b, 2k-a-b) \\ a+b+c \equiv 0 \pmod 2}} V_c$$

and

$$d_a = \dim_q(V_a) = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

Definition 15.3 ($SU(2)_k$ modular data)**.** For $0 \le a, b \le k$, define

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\!\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \qquad \theta_a = \exp\!\left(\frac{\pi i\, a(a+2)}{2(k+2)}\right).$$

Let ${N_{ab}}^c$ be the multiplicity of V_c in $V_a \otimes V_b$.

Exercise 15.2 (Verlinde diagonalization)**.** Use sine orthogonality to prove that S is real, symmetric, and unitary. For each a, let $(N_a)_{bc} = {N_{ab}}^c$. Use the truncated Chebyshev recursion to prove

$$N_a = S \; \mathrm{diag}\!\left(\frac{S_{ax}}{S_{0x}}\right)_{x=0}^{k} S^{-1}$$

and hence

$${N_{ab}}^c = \sum_{x=0}^{k} \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{0x}}.$$

Deduce that the number of fusion channels from $a_1, \dots, a_n$ to c is $(N_{a_1} \cdots N_{a_n})_{0c}$ and express it as a finite sum of products of sine functions.

Exercise 15.3 (Anyon interference and fusion counting)**.** For simple charges a, b, prove that the normalized monodromy amplitude is

$$M_{ab} = \frac{S_{ab} S_{00}}{S_{0a} S_{0b}}.$$

In a two-path anyon interferometer with path intensities I_1, I_2, derive

$$I_b(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2}\, \Re(e^{i\phi} M_{ab}).$$

Insert the sine formula and predict the visibility and phase for every enclosed charge. For a collection of prepared charges, use the Verlinde formula to predict the exact number of fusion outcomes in each total-charge sector. Identify indistinguishable charges for one probe and prove that the full family of probes distinguishes them. State the coherence, topological-gap, charge-preparation, and readout assumptions separately from the predicted fringes and integer state counts.

Part III

Counting

16 Graded Traces and Characters

Definition 16.1 (Symmetric powers and graded traces)**.** For a linear space V, define

$$\operatorname{Sym}^m V = V^{\otimes m}/\langle \xi - \sigma\xi : \xi \in V^{\otimes m},\ \sigma \in S_m \rangle.$$

An endomorphism T induces endomorphisms $\operatorname{Sym}^m T$ and $\Lambda^m T$. Their bosonic and fermionic graded traces are

$$Z_B(T; z) = \sum_{m \geq 0} \operatorname{tr}(\operatorname{Sym}^m T) z^m, \qquad Z_F(T; z) = \sum_{m \geq 0} \operatorname{tr}(\Lambda^m T) z^m.$$

Definition 16.2 (Partitions)**.** An *integer partition* of $n \geq 0$ is a finite nonincreasing sequence $\lambda = (\lambda_1, \dots, \lambda_r)$ of positive integers with $|\lambda| = \sum_i \lambda_i = n$. Let $p(n)$ be the number of partitions, with $p(0) = 1$.

Exercise 16.1 (Bosonic and fermionic trace generating functions)**.** Let V be a finite-dimensional complex linear space and $T \in \operatorname{End}(V)$. Using the intrinsic determinant defined by the action on the highest exterior power, prove

$$Z_B(T; z) = \frac{1}{\det(I - zT)}, \qquad Z_F(T; z) = \det(I + zT).$$

The first identity holds for $|z|$ below the reciprocal spectral radius; the second is a polynomial identity. Prove them first for a direct sum of generalized spectral subspaces and then by polynomial continuation, without choosing vectors in those subspaces.

Let the one-particle Hamiltonian have one-dimensional spectral subspaces at energies $j\hbar\omega$, $j \geq 1$. Apply the identities to the finite spectral truncation and then pass coefficientwise to the limit to derive

$$\prod_{j \geq 1} \frac{1}{1 - q^j} = \sum_{n \geq 0} p(n) q^n, \qquad \prod_{j \geq 1} (1 + q^j) = \sum_{n \geq 0} p_{\text{dist}}(n) q^n,$$

where $p_{\text{dist}}(n)$ counts partitions into distinct parts. Construct the multiplicity map separating powers of two and prove

$$\prod_{j \geq 1} (1 + q^j) = \prod_{j \geq 1} \frac{1}{1 - q^{2j-1}}.$$

Interpret the coefficient of q^n as the exact degeneracy at excitation energy $n\hbar\omega$ for the corresponding bosonic or fermionic system.

Definition 16.3 (Gaussian binomial coefficients)**.** For $0 \leq r \leq n$, define

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_r (q;q)_{n-r}},$$

and set it equal to zero outside this range.

Exercise 16.2 (Gaussian binomial theorem)**.** Derive

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \begin{bmatrix} n-1 \\ r \end{bmatrix}_q + q^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_q$$

and prove

$$\prod_{j=0}^{n-1} (1+tq^j) = \sum_{r=0}^{n} q^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_q t^r.$$

Prove that the coefficient of q^m in $\left[\begin{smallmatrix} n \\ r \end{smallmatrix}\right]_q$ counts partitions of m whose diagrams fit inside an $r \times (n-r)$ rectangle. For n equally spaced fermionic levels, identify the coefficient of $t^r q^m$ with the exact number of r-particle states of excitation m above the filled lowest configuration.

Exercise 16.3 (Hecke Poincaré identity)**.** Put $t = v^2$. For $w \in S_r$, let $\ell(w)$ be the least number of adjacent transpositions in an expression for w, and let $T_w = T_{i_1} \cdots T_{i_{\ell(w)}}$ for any reduced expression $w = s_{i_1} \cdots s_{i_{\ell(w)}}$. Prove from the braid relations that T_w is well-defined and that

$$\sum_{w \in S_r} t^{\ell(w)} = \prod_{j=1}^{r} \frac{1-t^j}{1-t}.$$

Prove that

$$e_+ = \frac{\sum_{w \in S_r} v^{\ell(w)} T_w}{\sum_{w \in S_r} v^{2\ell(w)}}, \qquad e_- = \frac{\sum_{w \in S_r} (-v^{-1})^{\ell(w)} T_w}{\sum_{w \in S_r} v^{-2\ell(w)}}$$

are idempotents satisfying $T_i e_+ = v e_+$ and $T_i e_- = -v^{-1} e_-$. Under the fundamental Hecke action from the preceding part, identify their images with the quantum symmetric and exterior state spaces and interpret the Poincaré polynomial as their graded state count.

Definition 16.4 (Virasoro characters)**.** The Virasoro algebra has generators L_n, $n \in \mathbb{Z}$, and a central element acting by c, with

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

A highest-weight vector $|h\rangle$ satisfies $L_0|h\rangle = h|h\rangle$ and $L_n|h\rangle = 0$ for $n > 0$. For τ in the upper half-plane put $q = e^{2\pi i \tau}$. If H_a is a positive-energy module with finite-dimensional L_0 spectral subspaces, define

$$\chi_a(\tau) = \mathrm{Tr}_{H_a}\, q^{L_0 - c/24}$$

when the displayed operator is trace class; the same expression denotes the formal graded trace otherwise.

Exercise 16.4 (Descendant trace)**.** Order the negative Virasoro modes by their indices. Use the commutation relations to rewrite every word as an ordered word. Check every three-letter overlap and use the Jacobi identity to prove that the rewriting is confluent. Filter by word length and deduce that the associated graded algebra is the symmetric algebra on the negative modes. Conclude that the Verma-module character is

$$\chi_{c,h}^{\mathrm{Verma}}(\tau) = \frac{q^{h-c/24}}{(q;q)_\infty}.$$

For the vacuum impose $L_{-1}|0\rangle = 0$ and derive the corresponding generic vacuum product. Show how a homogeneous null relation removes its entire descendant tower from the graded trace.

Definition 16.5 (Unitary modular character systems)**.** A finite character vector χ is a *unitary modular character system* when

$$\chi(-1/\tau) = S\chi(\tau), \qquad \chi(\tau+1) = T\chi(\tau)$$

for unitary matrices S, T. For a matrix M with nonnegative integral entries define

$$Z_M(\tau, \bar{\tau}) = \sum_{a,b} \overline{\chi_a(\tau)} M_{ab} \chi_b(\tau).$$

It is *modular invariant* when

$$S^* M S = M, \qquad T^* M T = M.$$

Exercise 16.5 (Ising characters and fusion counts)**.** For the Ising character system with $c = 1/2$ and weights $0, 1/2, 1/16$, put

$$\begin{aligned} \chi_0 &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}}\right), \\ \chi_{1/2} &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}}\right), \\ \chi_{1/16} &= \frac{1}{\sqrt{2}}\sqrt{\frac{\vartheta_2}{\eta}}. \end{aligned}$$

Use the theta and eta transformations developed in Fourier Analysis to derive

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \qquad T_{aa} = e^{2\pi i (h_a - c/24)}.$$

Expand each character as $q^{h-c/24}$ times a power series, identify its coefficients with exact descendant counts, and verify the modular invariance of $\sum_a |\chi_a|^2$. Apply the Verlinde formula from the root-of-unity section to recover the Ising fusion multiplicities.

Exercise 16.6 (Modular saddle-point counts)**.** Use Exercise 3.11 to prove

$$\prod_{j \geq 1} (1 - e^{-tj})^{-1} \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \qquad (t \downarrow 0).$$

Insert this estimate into Cauchy's coefficient integral, locate its positive real saddle, expand the phase through quadratic order, and bound the complementary contour to derive

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

For a modular-invariant unitary character system with unique vacuum, apply the same calculation to $Z_M(i\beta/(2\pi))$ and prove

$$\log \rho(N, \bar{N}) \sim 2\pi\sqrt{\frac{cN}{6}} + 2\pi\sqrt{\frac{\bar{c}\bar{N}}{6}}.$$

For the Ising system, derive the resulting asymptotic number of states below a specified high energy on a circle and the corresponding entropy. State the circumference, propagation speed, energy window, vacuum assumption, and finite-size limitations required when comparing these predictions with measured or transfer-matrix spectra.

17 Transfer-Matrix Traces and Finite-Size Spectra

Definition 17.1 (Transfer-matrix counts)**.** Let $\mathcal{R}_N$ be a finite set of allowed width-N configurations and let $T_N(z)$ be an operator on the configuration space whose matrix element between $a, b \in \mathcal{R}_N$ is the weight of adjoining b to a. For a height-M periodic lattice define

$$Z_{N,M}(z) = \mathrm{Tr}(T_N(z)^M).$$

When $T_N(z)$ has nonnegative entries, write $\lambda_N(z)$ for its spectral radius.

Definition 17.2 (Hard-hexagon rows)**.** Let

$$\mathcal{R}_N = \{a \in \{0,1\}^{\mathbb{Z}/N\mathbb{Z}} : a_i a_{i+1} = 0 \text{ for every } i\}.$$

Two rows $a, b \in \mathcal{R}_N$ are compatible when

$$a_i b_i = 0, \qquad a_i b_{i+1} = 0 \quad (i \in \mathbb{Z}/N\mathbb{Z}).$$

For $z \geq 0$, with $|a| = \sum_i a_i$, define

$$T_N(z)_{ab} = \mathbf{1}_{\{a,b \text{ compatible}\}} z^{(|a|+|b|)/2}.$$

Exercise 17.1 (Hard-hexagon trace counts)**.** Identify the horizontal, vertical, and diagonal nearest-neighbor edges encoded by the two compatibility conditions. Prove

$$\mathrm{Tr}(T_N(z)^M) = \sum_I z^{|I|},$$

where the sum is over independent vertex sets of the specified triangular $N \times M$ torus. Prove that every coefficient is an exact state count. For $z > 0$, apply the spectral-radius formula to show

$$\lim_{M\to\infty} \frac{1}{NM} \log Z_{N,M}(z) = \frac{1}{N} \log \lambda_N(z).$$

Derive the mean occupied fraction and finite-size susceptibility from the first two derivatives with respect to $\log z$. State the lattice geometry, boundary conditions, and finite N, M limitations required when these quantities are compared with adsorption or exclusion-process data.

Exercise 17.2 (Theta-character transfer spectra)**.** For a critical periodic transfer system of circumference L, define H_L by $T_L = e^{-aH_L}$ and take

$$H_L = E_{\text{bulk}} L + \frac{2\pi v}{L}\left(L_0 + \overline{L}_0 - \frac{c + \bar{c}}{24}\right), \qquad P_L = \frac{2\pi}{L}(L_0 - \overline{L}_0).$$

Use the Ising character system from the preceding section to prove that the coefficient of $q^{h+N-c/24}\bar{q}^{\bar{h}+\bar{N}-\bar{c}/24}$ is the multiplicity at

$$E - E_{\text{bulk}} L = \frac{2\pi v}{L}\left(h + \bar{h} + N + \bar{N} - \frac{c+\bar{c}}{24}\right), \quad P = \frac{2\pi}{L}(h - \bar{h} + N - \bar{N}).$$

Compute the first spectral gaps and degeneracies in the three Ising sectors. Deduce the universal gap ratios and the L^{-1} Casimir-energy correction. State how finite-size transfer eigenvalues determine c, the primary weights, and v, including the symmetry-sector resolution and irrelevant-operator corrections required for the comparison.

18 Determinantal Counting

Definition 18.1 (Probability spaces and convergence)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A random variable is a measurable map, and its law is its pushforward of $\mathbb{P}$. The expectation of an integrable scalar random variable is its integral. A sequence X_N converges in probability to X when

$$\mathbb{P}(d(X_N, X) > \varepsilon) \longrightarrow 0 \qquad (\varepsilon > 0),$$

and converges in distribution when its laws converge against bounded continuous functions.

Exercise 18.1 (Hermite Gaussian moments)**.** For $X \sim N(0,1)$, prove

$$\mathbb{E}[e^{tX}] = e^{t^2/2}.$$

Define the probabilists' Hermite polynomials by

$$e^{xt - t^2/2} = \sum_{n \geq 0} \mathrm{He}_n(x) \frac{t^n}{n!}$$

and prove

$$\mathbb{E}[\mathrm{He}_m(X)\,\mathrm{He}_n(X)] = n!\delta_{mn}, \qquad \mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n n!}.$$

Relate He_n to the oscillator Hermite functions developed in the Kirillov section.

Definition 18.2 (Dyson classes)**.** For $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, put $\beta = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}$. The Gaussian Dyson ensemble on the self-adjoint operators on $\mathbb{F}^N$ has density

$$d\mathbb{P}_{N,\beta}(H) = C_{N,\beta} \exp\left[-\frac{\beta N}{4\sigma^2} \operatorname{Tr}(H^2)\right] dH$$

for the invariant Hilbert–Schmidt volume. For $\lambda = (\lambda_1, \ldots, \lambda_N)$ define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^N.$$

Exercise 18.2 (Orbit Jacobian and level repulsion)**.** Use the exterior-power determinant to prove

$$\Delta(\lambda) = \prod_{i<j} (\lambda_j - \lambda_i).$$

For each Dyson class, compute the Jacobian of the conjugation-orbit map on the regular spectral locus with the stabilizer and permutation factors included, and derive

$$p_{N,\beta}(\lambda) = \frac{1}{Z_{N,\beta}} e^{-\beta N \sum_i \lambda_i^2/(4\sigma^2)} |\Delta(\lambda)|^\beta.$$

For 2×2 GOE and GUE, separate the trace from the traceless operator, rescale the gap to unit mean, and obtain

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \qquad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Identify the exponent β as the small-gap repulsion prediction and state the symmetry-sector separation required in spectral data.

Definition 18.3 (Fredholm determinants)**.** For $A \in S_1(H)$ define

$$\det(I + zA) = \prod_j (1 + z\lambda_j(A)),$$

where the nonzero eigenvalues are repeated with algebraic multiplicity.

Exercise 18.3 (Fredholm trace identity)**.** Prove absolute local uniform convergence of the defining product and

$$\det(I - zA) = \exp\left(-\sum_{m\geq 1} \frac{z^m}{m} \operatorname{Tr}(A^m)\right) \qquad (|z| < \|A\|^{-1}).$$

Prove that the determinant is entire, identify its zeros, and show that its finite-rank specialization agrees with the exterior-power identity from the opening section.

Definition 18.4 (Point processes and correlation densities)**.** Let (X, μ) be locally compact. A simple point process is a random locally finite counting measure $\Xi = \sum_i \delta_{x_i}$ with distinct points almost surely. When its kth factorial moment measure is absolutely continuous with respect to $\mu^{\otimes k}$, its density ρ_k is defined by

$$\mathbb{E} \sum_{i_1,\ldots,i_k}^{\neq} f(x_{i_1}, \ldots, x_{i_k}) = \int_{X^k} f \rho_k \, d\mu^{\otimes k}.$$

A *determinantal point process* with kernel K is one for which

$$\rho_k(x_1, \ldots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^k.$$

For the existence and counting formulas below, K is a locally trace-class positive contraction on $L^2(X, \mu)$.

Exercise 18.4 (Determinantal counting statistics)**.** For a relatively compact measurable set B, put $N_B = \Xi(B)$ and $K_B = \mathbf{1}_B K \mathbf{1}_B$. Use factorial moments and the Fredholm series to prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z-1)K_B), \qquad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Prove that N_B is distributed as a sum of independent Bernoulli variables whose parameters are the eigenvalues of K_B.

Definition 18.5 (Unitary polynomial ensembles)**.** Let $w(x)d\mu(x)$ be a positive measure with finite moments and let p_j be its monic orthogonal polynomials, normalized by

$$\int p_j(x) p_k(x) w(x) d\mu(x) = h_j \delta_{jk}.$$

The associated unitary polynomial ensemble has density

$$p_N(x_1, \ldots, x_N) = \frac{1}{N! \prod_{j=0}^{N-1} h_j} \Delta(x)^2 \prod_{i=1}^N w(x_i)$$

and kernel

$$K_N(x, y) = \sqrt{w(x)w(y)} \sum_{j=0}^{N-1} \frac{p_j(x)p_j(y)}{h_j}.$$

Exercise 18.5 (Andréief and Christoffel–Darboux identities)**.** Prove Andréief's identity

$$\int_{X^N} \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu(x_j) = N! \det\left[\int_X f_i g_j \, d\mu\right].$$

Deduce that the unitary polynomial ensemble is determinantal with kernel K_N. Derive the Christoffel–Darboux formula from the three-term recurrence and prove

$$\int K_N(x,z)K_N(z,y)d\mu(z) = K_N(x,y), \qquad \int K_N(x,x)d\mu(x) = N.$$

Definition 18.6 (GUE and its limiting kernels)**.** For the GUE weight $w_N(x) = e^{-Nx^2/(2\sigma^2)}$ and the normalized oscillator Hermite functions h_j, define

$$\phi_{j,N}(x) = \left(\frac{N}{2\sigma^2}\right)^{1/4} h_j\left(\sqrt{\frac{N}{2\sigma^2}}x\right), \qquad K_N^{\mathrm{GUE}}(x,y) = \sum_{j=0}^{N-1} \phi_{j,N}(x)\phi_{j,N}(y).$$

Define

$$\rho_{\mathrm{sc}}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2 - x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x),$$

$$K_{\sin}(u,v) = \frac{\sin \pi(u-v)}{\pi(u-v)},$$

$$K_{\mathrm{Ai}}(u,v) = \frac{\mathrm{Ai}(u)\,\mathrm{Ai}'(v) - \mathrm{Ai}'(u)\,\mathrm{Ai}(v)}{u-v}, \qquad F_2(s) = \det(I - K_{\mathrm{Ai}})_{L^2(s,\infty)}.$$

Diagonal values are defined by continuity.

Exercise 18.6 (Hermite bulk and turning-point limits)**.** Derive a contour integral for the normalized Hermite functions from their generating function. For $E \in (-2\sigma, 2\sigma)$, perform steepest descent at the two simple saddles; at 2σ, rescale the coalescing saddles to the Airy integral. Establish uniform bounds on compact bulk windows and exponentially weighted edge half-lines sufficient to prove

$$\frac{1}{N\rho_{\mathrm{sc}}(E)} K_N^{\mathrm{GUE}}\left(E + \frac{u}{N\rho_{\mathrm{sc}}(E)}, E + \frac{v}{N\rho_{\mathrm{sc}}(E)}\right) \longrightarrow K_{\sin}(u,v),$$

$$\frac{\sigma}{N^{2/3}} K_N^{\mathrm{GUE}}\left(2\sigma + \frac{\sigma u}{N^{2/3}}, 2\sigma + \frac{\sigma v}{N^{2/3}}\right) \longrightarrow K_{\mathrm{Ai}}(u,v).$$

Use the uniform bounds to pass to the Fredholm series. Deduce the GUE largest-eigenvalue law

$$\mathbb{P}\left(\frac{N^{2/3}}{\sigma}(\lambda_{\max} - 2\sigma) \le s\right) \longrightarrow F_2(s)$$

and derive the logarithmic number variance of a long interval from $K_{\sin}$. State the unfolding, centering, symmetry class, independence, and finite-size assumptions required when testing these predictions with resonance or covariance spectra.

19 Noncrossing Trace Limits

Definition 19.1 (Wigner matrices)**.** A Wigner matrix H_N is an $N \times N$ Hermitian random matrix whose upper-triangular entries are independent and centered, with $\mathbb{E}|H_{ij}|^2 = \sigma^2/N$ off the diagonal and uniformly bounded rescaled moments of every fixed order. Its empirical spectral measure is

$$L_{H_N} = \frac{1}{N}\sum_j \delta_{\lambda_j(H_N)}, \qquad \int x^k \, dL_{H_N}(x) = \frac{1}{N}\operatorname{Tr}(H_N^k).$$

Definition 19.2 (Free cumulants)**.** A noncommutative probability space is a unital complex algebra A with a unital linear functional φ. In a $*$-algebra, a state is positive; it is tracial when $\varphi(ab) = \varphi(ba)$. Let $NC(n)$ be the lattice of noncrossing partitions of $\{1, \dots, n\}$. The free cumulants are the multilinear maps determined by

$$\varphi(a_1 \cdots a_n) = \sum_{\pi \in NC(n)} \kappa_\pi(a_1, \dots, a_n).$$

Unital subalgebras are freely independent when alternating products of centered elements from neighboring distinct subalgebras have zero expectation.

Definition 19.3 (Cauchy and R-transforms)**.** For a compactly supported probability measure μ on $\mathbb{R}$, define

$$G_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(t)}{z - t}.$$

Near infinity let $K_\mu = G_\mu^{\langle -1 \rangle}$ and define

$$K_\mu(w) = \frac{1}{w} + R_\mu(w).$$

The free additive convolution $\mu \boxplus \nu$ is the law of $X + Y$ when X, Y are freely independent with laws μ, ν.

Exercise 19.1 (Semicircle equivalences)**.** Expand $N^{-1}\mathbb{E}\operatorname{Tr}(H_N^k)$ as a sum over closed index walks and prove that the leading pair matches are indexed by noncrossing pairings. Deduce

$$\lim_{N\to\infty} \frac{1}{N}\mathbb{E}\operatorname{Tr}(H_N^{2m}) = C_m \sigma^{2m}, \qquad C_m = \frac{1}{m+1}\binom{2m}{m},$$

with odd moments zero, and prove that the variance tends to zero. Decompose a noncrossing pairing at the partner of its first point to obtain

$$C_{m+1} = \sum_{j=0}^{m} C_j C_{m-j}.$$

Use this recurrence to prove

$$\sigma^2 G(z)^2 - zG(z) + 1 = 0, \qquad zG(z) \longrightarrow 1,$$

and recover ρ_{sc} by Stieltjes inversion.

Show from the moment–cumulant relation that this law has only $\kappa_2 = \sigma^2$ nonzero. Finally, minimize

$$\mathcal{I}(\mu) = \frac{1}{2\sigma^2}\int x^2 d\mu(x) - \iint \log|x - y| \, d\mu(x) d\mu(y)$$

and prove that its Euler–Lagrange equation gives the same density. Thus identify the logarithmic equilibrium law, Catalan trace moments, quadratic Cauchy equation, and free Gaussian law as four formulations of the same limit.

Exercise 19.2 (Free central limit and convolution)**.** Prove from the alternating centered-product definition that mixed free cumulants vanish. Deduce

$$R_{\mu\boxplus\nu} = R_\mu + R_\nu.$$

For freely independent identically distributed centered self-adjoint variables of variance σ^2, prove by cumulant scaling that their normalized sums converge in moments to the semicircle law. Show that the free sum of semicircle laws of variances σ_1^2, σ_2^2 is semicircular of variance $\sigma_1^2 + \sigma_2^2$.

Definition 19.4 (Asymptotic freeness)**.** Random matrix families $A_N^{(1)}, \ldots, A_N^{(r)}$ are asymptotically free when

$$\lim_{N\to\infty} \frac{1}{N} \mathbb{E}\,\mathrm{Tr}\, P(A_N^{(1)}, \ldots, A_N^{(r)}) = \varphi(P(a_1, \ldots, a_r))$$

for every noncommutative polynomial P, where the limiting variables are freely independent.

Exercise 19.3 (Free spectral addition)**.** For independent GUE matrices, apply Wick contraction to normalized mixed traces and prove that crossing pairings are lower order. Deduce asymptotic freeness. For two independently coupled large subsystems with limiting semicircle variances σ_1^2 and σ_2^2, derive

$$G(z) = \frac{z - \sqrt{z^2 - 4(\sigma_1^2 + \sigma_2^2)}}{2(\sigma_1^2 + \sigma_2^2)}$$

and predict the density-of-states edges $\pm 2\sqrt{\sigma_1^2 + \sigma_2^2}$. State how empirical Stieltjes transforms and observed spectral edges test the freeness and large-size assumptions.

20 Schur–Macdonald Measures and Fluctuations

Definition 20.1 (Row insertion and RSK)**.** A semistandard Young tableau has weakly increasing rows and strictly increasing columns. To insert an integer x, replace the first entry strictly larger than x in the first row and insert the replaced entry in the next row; append when no such entry exists. For $A = (a_{ij}) \in M_{m,n}(\mathbb{Z}_{\geq 0})$, form the lexicographically ordered two-line word containing a_{ij} copies of (i, j), insert the lower letters, and record the upper letters in the created boxes. The resulting pair (P, Q) of tableaux of common shape λ is the RSK image of A.

Definition 20.2 (Last-passage values)**.** For nonnegative weights $A = (a_{ij})$, define

$$G(m, n) = \max_{\pi:(1,1)\nearrow(m,n)} \sum_{(i,j)\in\pi} a_{ij}.$$

For parameters $a_i, b_j \geq 0$ with $a_i b_j < 1$, the geometric model has independent weights

$$\mathbb{P}(a_{ij} = k) = (1 - a_i b_j)(a_i b_j)^k, \qquad k \in \mathbb{Z}_{\geq 0}.$$

Exercise 20.1 (RSK and the finite Schur measure)**.** Prove that row insertion is reversible and hence that RSK is a bijection. Show that the length of the first row of its shape is the maximal weight of an up-right path:

$$\lambda_1 = G(m, n).$$

For variables $a = (a_1, \ldots, a_m)$, define

$$s_\lambda(a) = \sum_T \prod_i a_i^{m_i(T)}.$$

Use the contents of the two RSK tableaux and the geometric weight law to derive

$$\mathbb{P}(\lambda) = \prod_{i,j}(1 - a_i b_j) s_\lambda(a) s_\lambda(b)$$

and the Cauchy identity

$$\sum_\lambda s_\lambda(a) s_\lambda(b) = \prod_{i,j}(1 - a_i b_j)^{-1}.$$

Definition 20.3 (Finite Schur kernel)**.** For the preceding finite alphabets, set

$$\Phi(z) = \frac{\prod_j (1 - b_j/z)}{\prod_i (1 - a_i z)}.$$

Choose concentric contours with

$$\max_j b_j < |w| < |z| < \min_i a_i^{-1}.$$

For $r, s \in \mathbb{Z} + \frac{1}{2}$, define

$$K_{a,b}(r,s) = \frac{1}{(2\pi i)^2} \oint\oint \frac{\Phi(z)}{\Phi(w)} \frac{1}{z - w} z^{-r-1/2} w^{s-1/2}\, dz\, dw.$$

Exercise 20.2 (Schur determinantal counts)**.** Encode λ by

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \tfrac{1}{2} : i \geq 1\}.$$

Use the Cauchy identity and coefficient extraction on the displayed contours to prove that $\mathcal{X}(\lambda)$ is determinantal with kernel $K_{a,b}$. Deduce the exact finite Fredholm formula

$$\mathbb{P}(G(m,n) \leq L) = \det(I - K_{a,b})_{\ell^2(\{L+\frac{1}{2}, L+\frac{3}{2}, \ldots\})}.$$

State how repeated last-passage or growth experiments with the specified geometric weights test this distribution without invoking an asymptotic limit.

Definition 20.4 (Finite Macdonald measures)**.** Let $0 \leq q, t < 1$, let $a = (a_1, \ldots, a_N)$ and $b = (b_1, \ldots, b_M)$ be nonnegative with $a_i b_j < 1$, and let P_λ be the Macdonald polynomials developed in the DAHA section. Let $Q_\lambda = b_\lambda(q,t) P_\lambda$ be the family dual to P_λ for the Macdonald inner product. Define

$$\Pi(a;b) = \prod_{i,j} \frac{(t a_i b_j; q)_\infty}{(a_i b_j; q)_\infty}, \qquad \mathbb{P}_{q,t}(\lambda) = \frac{P_\lambda(a;q,t) Q_\lambda(b;q,t)}{\Pi(a;b)}.$$

Exercise 20.3 (One Macdonald observable)**.** Apply the first Macdonald difference operator in the a variables to the Cauchy identity. For distinct a_i, prove

$$\mathbb{E}_{q,t}\left[\sum_{i=1}^N q^{\lambda_i} t^{N-i}\right] = \sum_{i=1}^N \prod_{j \neq i} \frac{t a_i - a_j}{a_i - a_j} \prod_{k=1}^M \frac{1 - a_i b_k}{1 - t a_i b_k},$$

and extend the identity to coincident parameters by continuity. Verify the Schur specialization $q = t$ and the q-Whittaker specialization $t = 0$. For samples from the finite measure, identify the displayed expectation as a directly estimable statistic and state the positivity, truncation, and sampling assumptions required for comparison with the formula.

Exercise 20.4 (Laguerre last-passage fluctuations)**.** Set $a_i = b_j = e^{-\varepsilon/2}$ in the $n \times n$ geometric model and prove that εa_{ij} converges in distribution to a rate-one exponential variable as $\varepsilon \downarrow 0$. Take this limit in the finite Schur Fredholm formula and identify the resulting kernel with the Laguerre Christoffel–Darboux kernel. Use the Laguerre contour integral and a double-saddle expansion at the upper spectral edge, together with uniform exponential bounds for the Fredholm series, to prove

$$\frac{G(n,n) - 4n}{2^{4/3} n^{1/3}} \Longrightarrow \chi_2, \qquad \mathbb{P}(\chi_2 \leq s) = F_2(s).$$

Deduce the $n^{1/3}$ height-fluctuation scale and use the quadratic curvature of the deterministic limit shape to obtain the transverse $n^{2/3}$ scale. State the independent-weight, boundary, centering, scale-calibration, and finite-size assumptions required when the Tracy–Widom prediction is compared with interface-height or directed transport data.